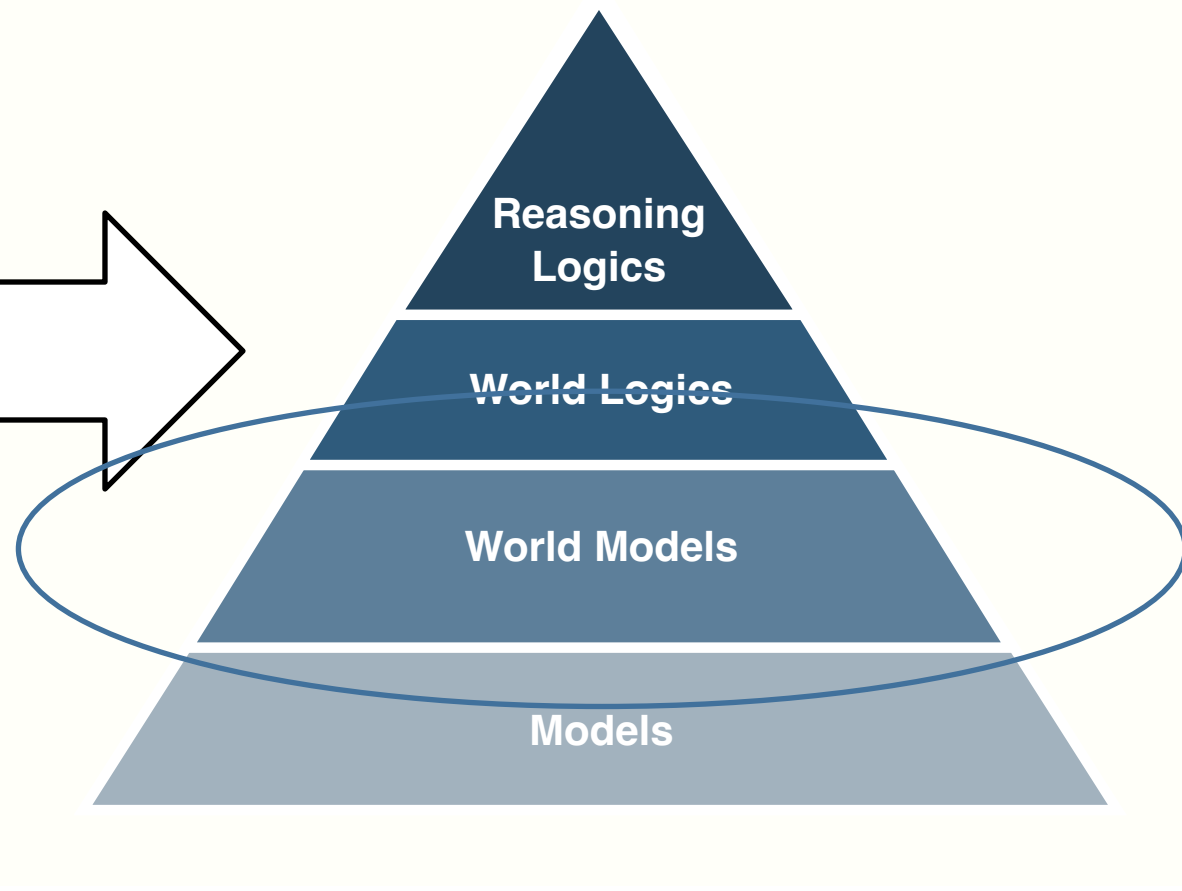
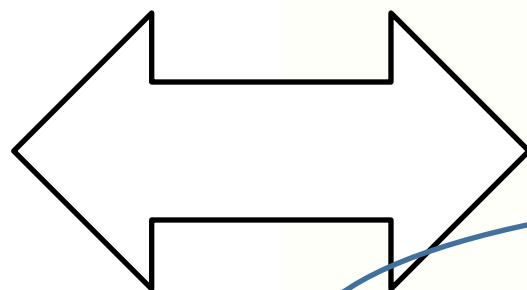
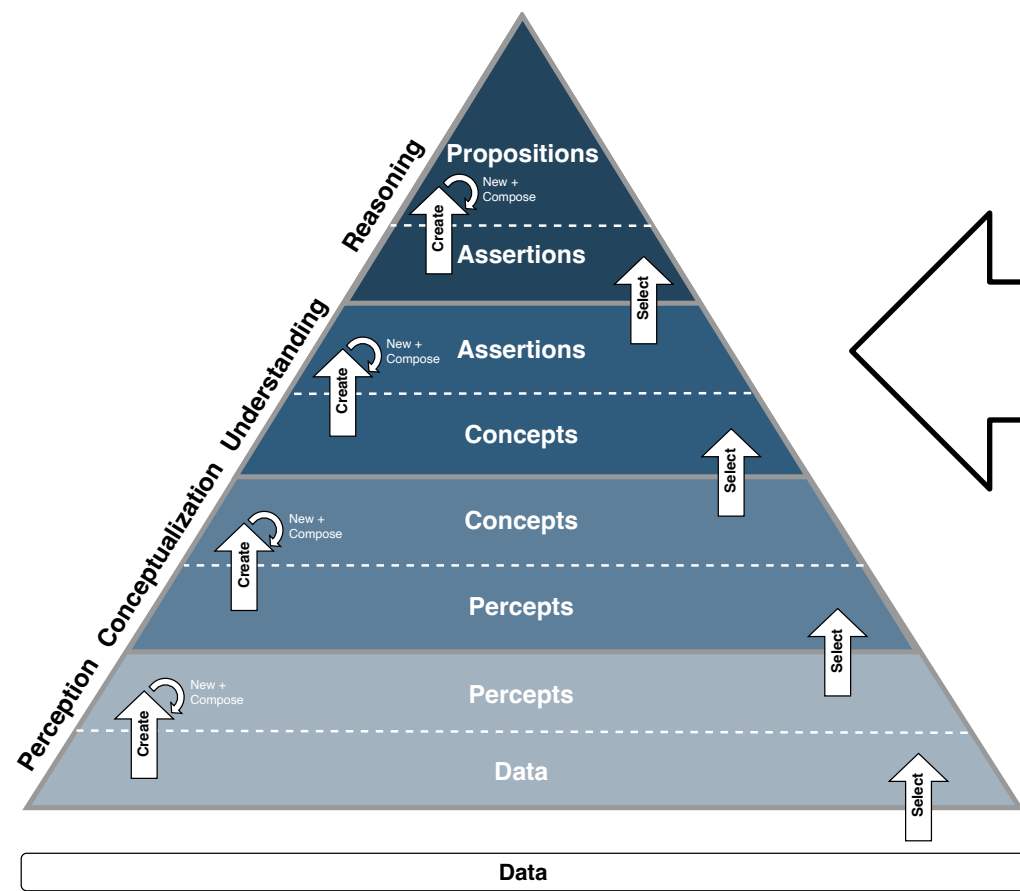




World model Types (HP2T)

The knowledge Pyramid and the Logics pyramid (recall)



World models – basic and composed types

Three basic types of world models (**base** world models)

- Concept graphs (CG)
- Etype (entity type) graphs (ETG)
- Entity graphs (EG)

Four types of compositions of world models (**composed** world models)

- Concept + etype graphs
- Concept + entity graphs
- Etype + entity graphs
- Concept + etype + entity graphs

Index

- **Concept graphs (CG)**
- Etype (entity type) graphs (ETG)
- Entity types and data types
- Entity graphs (EG)
- Concept + etype graphs
- Concept + entity graphs
- Etype + entity graphs
- Concept + etype + entity graphs
- Semi-formal world models
- Key notions

World models as concept graphs

Concept graphs are linguistic representations of analogical representations which describe

- Concepts (as in «woman», «man», «person»)
- «**concept inclusion**» (as in «a woman is a person») and «**concept exclusion**» («**concept disjointness**») (as in «a woman is not a man») relations between two concepts

Reminder (concepts and percepts). Concepts denote sets of percepts... with the possibility of:

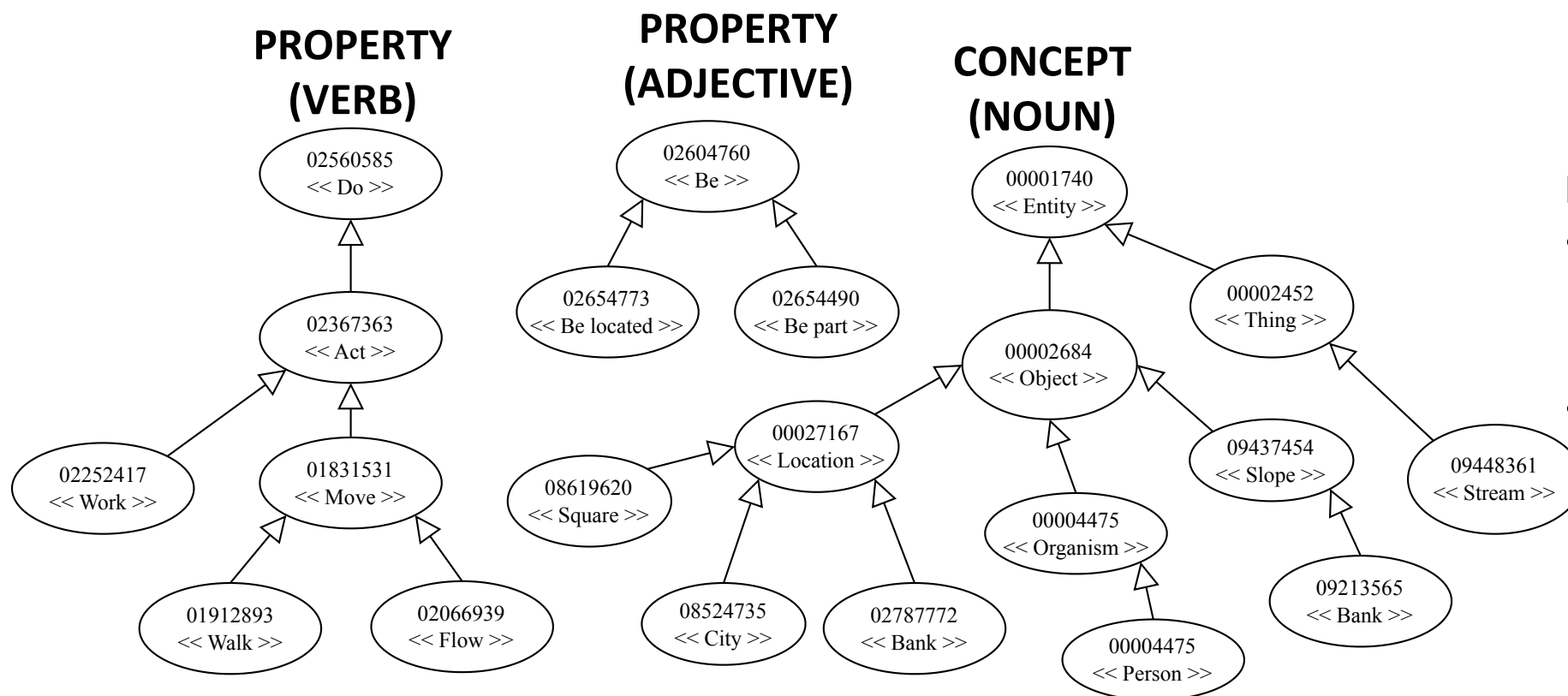
- two (sets of) percepts, as denoted by concepts, to be in different places at the same time
- two (sets of) percepts, as denoted by concepts, to be perceived at different times
- two (sets of) percepts, as denoted by concepts, to be perceived in an arbitrarily chosen (but fixed) period of time

Concept graphs

Concept graphs take the form of hierarchies where, semantically

- The root node is a concept which denotes the **domain of interpretation**
- A child node denotes a subset of the extension of the parent node (that is, we have **concept inclusion**),
- Any two sibling nodes (i.e., children of the same parent node) are pairwise disjoint (that is, we have **concept exclusion**)

Concept graphs - example



NOTE:

- All siblings are pairwise disjoint, e.g.,
 $I(\text{organism}) \cap I(\text{location}) = \emptyset$
- All children are subsets of the parents, e.g.,
 $I(\text{city}) \subseteq I(\text{location})$

Concept graphs

Terminology (Language models). Concept graphs are also called **language models** as they unambiguously **define** the meaning of the terms used in (natural / formal) language.

Observation (Language models). Language models allow to create a one-to-one mapping between what we perceive (our analogical representations) and the words (i.e., nouns, verbs, adjectives and adverbs) we use.

Terminology (Ontologies). Concept graphs which describe the world as it is / appears are called (in the Artificial Intelligence literature) **ontologies**. They are also called **top level ontologies**.

Terminology (Isa, partof). In ontologies, concept inclusion links are often called **isa (is-a)** links, as in «woman isa person». Some ontologies also have **partof (part-of)** links (as in «leg part-of body») – not considered in this course.

Observation (Language teleontologies, LTLO, domain languages). It is possible to build concept graphs focused on **specific** aspects of the world. We call this type of concept graphs, **language teleontologies (LTLO)**. This leads to the construction of so-called **domain LTLO** (used in domains such as, e.g., health, spacetime), **task LTLO** (used in tasks such as, e.g., industrial assembly), **application LTLO** (within a specific domain, e.g., health used in specific applications, e.g., hearth surgery). Given their importance and extensive use, domain LTLO are also called **domain languages**.

Index

- Concept graphs (CG)
- **Entity types and data types**
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- Entity graphs (EG)
- Concept + etype graphs
- Concept + entity graphs
- Etype + entity graphs
- Concept + etype + entity graphs
- Semi-formal world models
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Analogic representations – Entities (recall)

Observation (Entities). We have the following:

- **Entities**, that is, what we perceive from **things**, e.g., Sofia, a dog;
- **Entity types**, also called **etypes**, that is, **multiple entities which somehow look the same**, e.g., the etype woman, dog, tree;
- **Entity properties**, that is, **entities with certain characteristics**, e.g., a woman with blond hair, a dogs barking;
- **Property types**, that is, **multiple entities with certain characteristics**, similar among them, e.g., two women with blond hair, two dogs barking;
- **Entity Relations**, that, is **multiple entities in some relation among them**, e.g., a woman talking to a man with a phone, a dog between a woman and a man, a woman friend of a man;
- **Relation types**, that is, **groups of multiple entities in some relation among them**, similar among them, e.g., two groups of a man and woman talking to one another with a phone, two couples woman and man, both with a dog in between.



Relations and properties are **identified after the identification** of the entity: Do you see a woman which is black (together with a white one) or a black woman?

Etypes - examples

Location, i.e., an example etype which models spatial containment of entities.

Locations do not change their position with respect to their coordinate reference systems. That is, coordinates are properties of locations. They are key for deciding whether two locations (i.e., two entities belonging to the etype Location) are actually the same location.

There are many etypes which are special cases (sub-etypes) of Location, for instance: *Mountain, City, Street, Home*, and many others.

Other important etypes are:

Entity, the most general etype, that which contains all entities, as used within EGs. Its most relevant property is that it has a **unique identifier**, thus imposing that all entities must have it. The identifier is the identifying data property. In most cases it does not exist;

Event, whose most characterizing properties are its start and end times. Which identifying data properties?

Person, with properties such as name, birth date, parents, phone number; and many others. Which identifying data properties?

Entity types and Data types

Definition (Data type, value). A **data type (dtype)** is an **etype** whose properties are defined a priori. A dtype is defined by providing: (i) the names of its values plus the interpretation function; (ii) n-ary properties and their values (iii) the notion of equality between values

Example (Data type). Number, real, string, identifier, date, time, coordinates, weight, data (this being the most general datatype) polygon, point, sequence of lines.

Observation (Data type, data value). Data types must be used “as is”. The use of data types is a well established practice in Computer Science. LoE allows much richer data types (e.g., date, geodatypes, mood data types) as needed to model the world. Value identification is guaranteed via equality.

Intuition (data type). Data types are etypes used to describe linguistically real world entities, built out of global modelling consensus.

Intuition (Object properties and data properties). **Object properties**, previously called **entity relations**, are property types whose codomain is an etype. **Data properties**, previously called **entity properties**, are property types whose codomain is a dtype.

Data types vs. Entity types – example

Example (Natural Number). We have the following

- Values: numerals, written 1,2,3, ..., denoting natural numbers, 1,2,3,...
- Unary properties: odd, even, prime
- Binary properties: plus, minus, times, division
- Equality: =

Example (Person). We have the following *user-defined* example:

- Entities: Sofia, Anonymus, Marco, Fausto, Pietro ... *denoting* Sofia#3, #1211, Marco#7, Fausto#1, ...
- Data properties: name, height, age, gender, coordinates, ...
- Object properties: near, friendOf, hasFriend, talksTo, ...
- Equality: FaceRecognition, SocialSecurityNumber, ...

Index

- Concept graphs (CG)
- Entity types and data types
- **Etype (entity type) graphs (ETG)**
- Entity graphs (EG)
- Concept + etype graphs
- Concept + entity graphs
- Etype + entity graphs
- Concept + etype + entity graphs
- Semi-formal world models
- Key notions

Etype graphs

Etype graphs take the form of labeled directed graphs where, semantically

- The root node (etype) is an etype which has the the same extension of a reference concept which defines the Universe of interpretation for the type model (as in «person»)
- An etype which is not the root etype can be defined by a **description** which associates a set of properties to a **MG etype**, thus making it a **LG etype** (as in «a bachelor is a married person»)
- An etype which is not the root etype can be defined as having a LG/MG relation with a **MG/LG etype** parent node, thus making it a **LG/MG etype** (as in «a bachelor is a person»»)

World models as etype graphs

Etype graphs are linguistic representations of analogical representations which describe

- Etypes (as in «married person»)
- Properties, denoting relations, between etypes (as in «friend of «has friend»)

where: etypes and properties denote subsets of the percepts and pairs or percepts, respectively, as denoted by a **reference concept** which defines the universe of interpretation

NOTE: etype graphs may or may not have «**more/less general (MG/LG) etypes**» or **etype disjointness**» relations denoting, respectively, subset and empty intersection.

Etype graphs

Terminology (knowledge models). Etype graphs are also called **knowledge models** as they unambiguously **describe** the properties which allow to distinguish between different occurrences of the root etype / concept.

Observation (knowledge models). Once the reference concept (and corresponding language model) unambiguously defines how the domain of interpretation is organized, knowledge models allow us to differentiate the different occurrences of the same concept based on their specific properties.

Observation (Concept and etype graphs). An etype graph can contain multiple concepts as long as the etype MG/LG hierarchy follows the concept inclusion hierarchy of the reference concept graph

Observation (Concept and etype). When talking we sometimes use the same word, e.g., «person» to denote a concept (as in «a person is not an animal») or an etype (as in «persons are friends of animals»). The default understanding is that this is the etype associated to the corresponding reference concept.

Etype graphs – Types of knowledge models

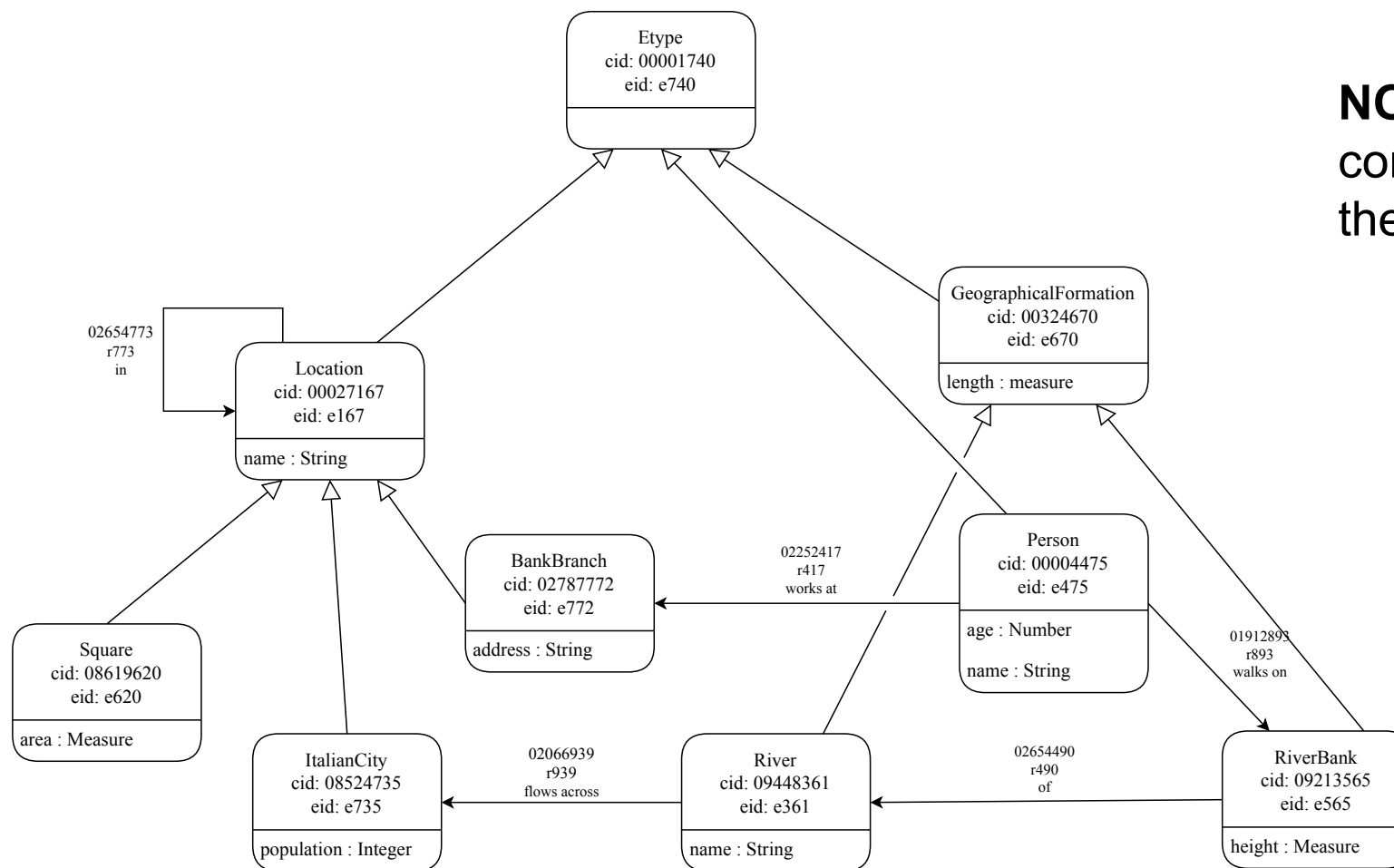
Terminology (Knowledge ontologies). Knowledge ontologies are knowledge models constructed from ontologies where one or more concepts are refined into etypes based on their properties.

Terminology (Teleontologies, taxonomies). Teleontologies (KTLO) are knowledge ontologies with one or more MG/LG relations. Taxonomies (KTAX) are tree-like KTLO where the extension of the union of the children is the same as the extension of the parent node.

Observation (knowledge ontologies). Similarly to language models and ontologies, Knowledge ontologies are distinguished based on the types of concepts and the generality of the top level concept (e.g., **top level KTLO**, **domain KTLO**, **task KTLO** and **application KTLO**).

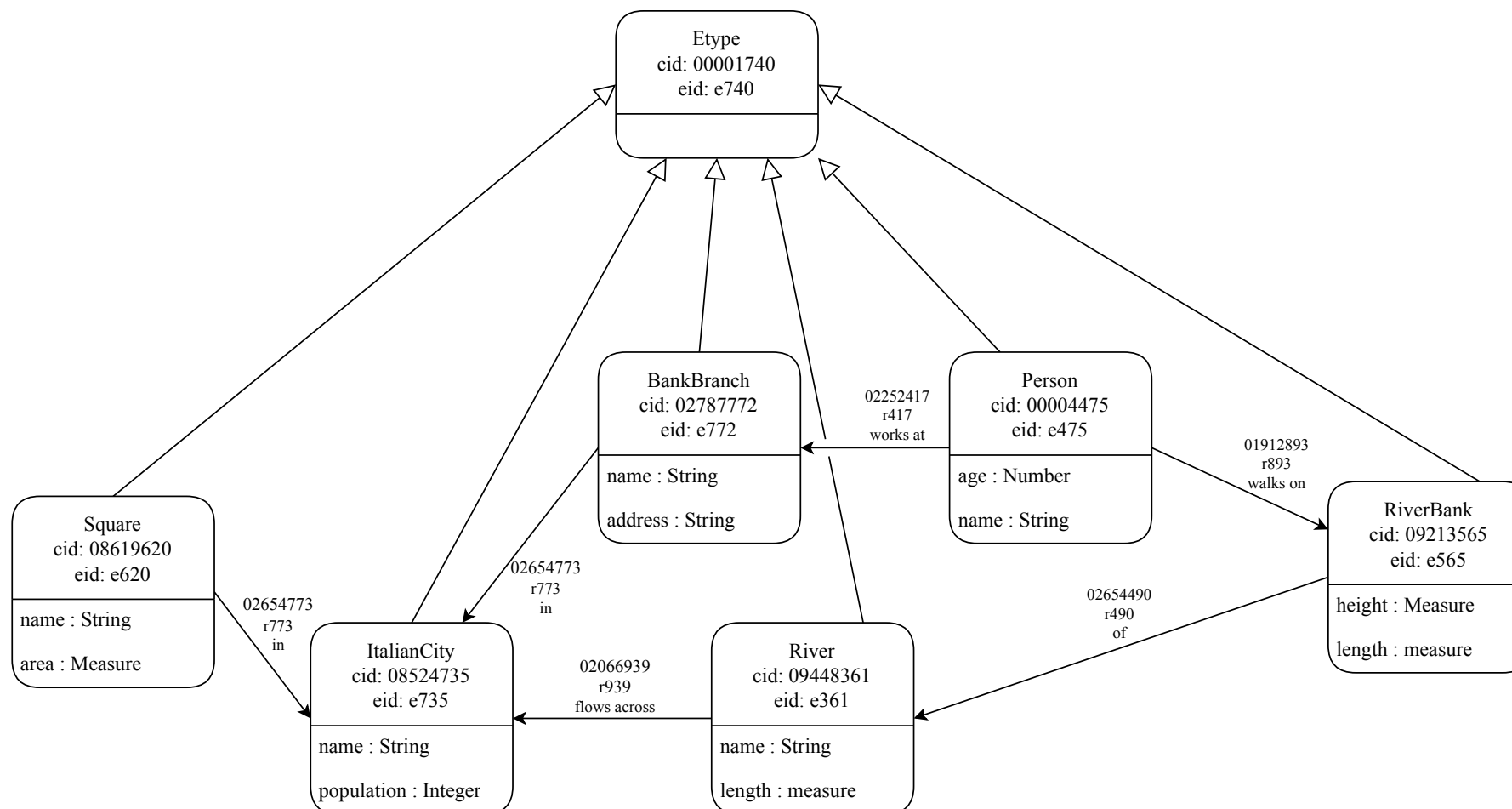
Terminology (Teleologies, schemas). Teleologies (TLO) are KTLO with MG/LG relation only to the root etype (i.e., reference concept). **Schema Graphs (SG)**, also called **schemas**, are teleologies where the root etype is left implicit.

Knowledge teleontology (KTLO) – example

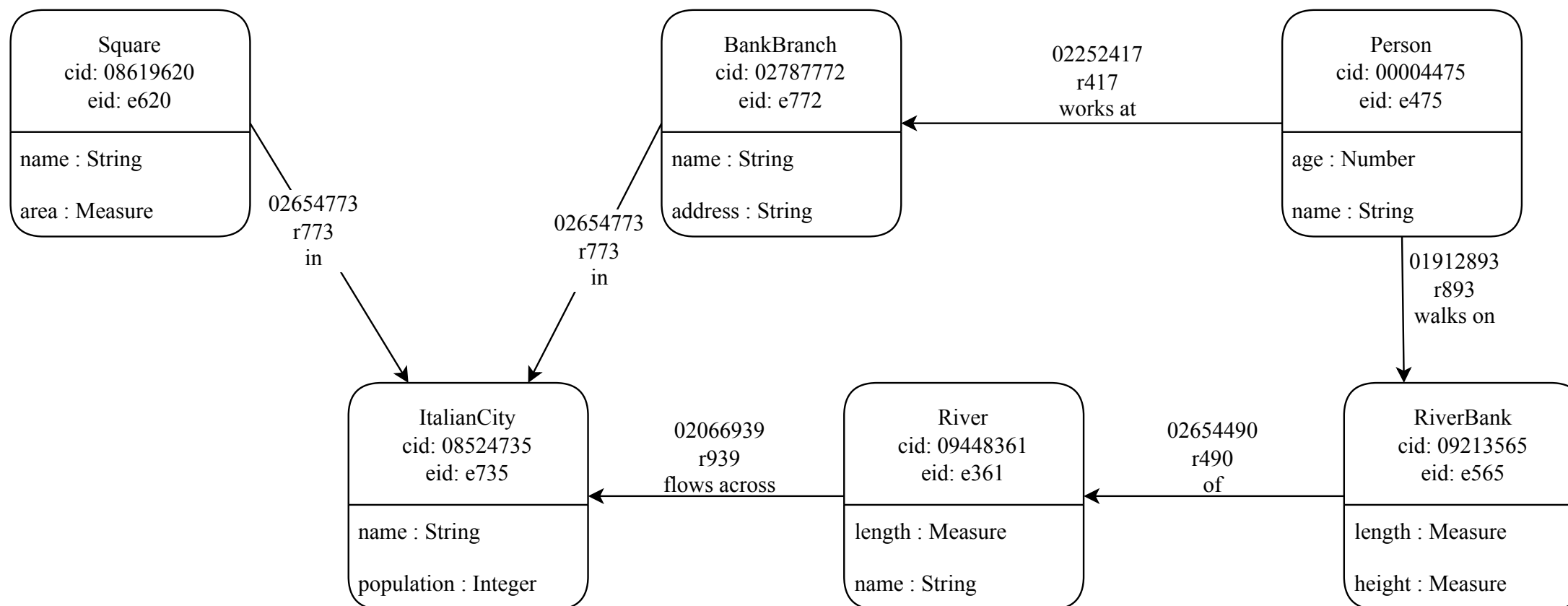


NOTE: Cid's of types connect to the Cid of the reference concept

Teleology (TLO) – example



Schema (SG) – example



Index

- Concept graphs (CG)
- Entity types and data types
- Etype (entity type) graphs (ETG)
- **Entity graphs (EG)**
- Concept + etype graphs
- Concept + entity graphs
- Etype + entity graphs
- Concept + etype + entity graphs
- Semi-formal world models
- Key notions

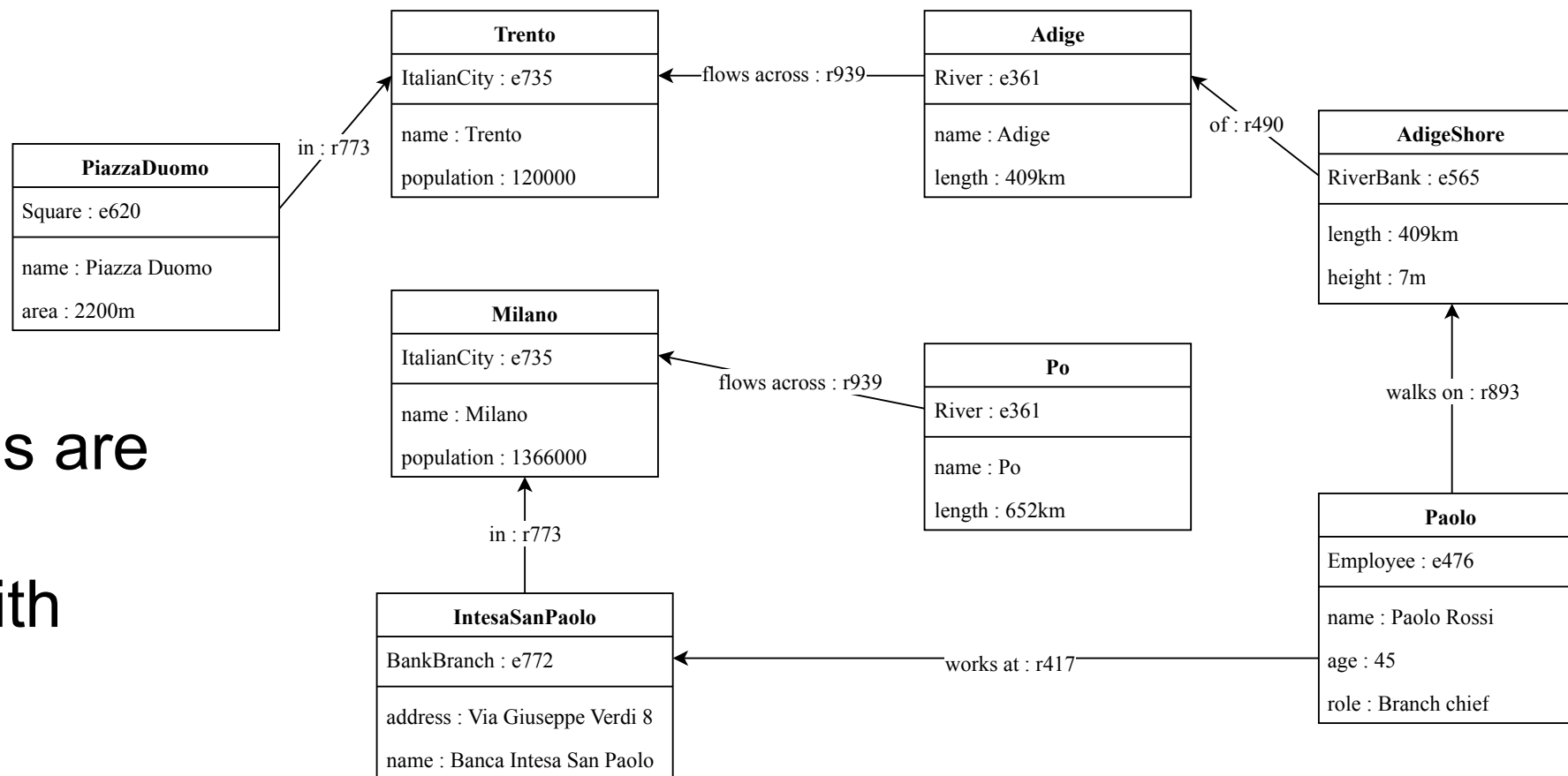
World models as entity graphs

Entity graphs are linguistic representations of analogical representations which describe

- entities (as in «Trento», «Duomo»)
- relations between entities (as in «friend of», «has friend»)

where: entities and relations denote members of the etypes and relations between etypes, as represented by a **reference schema**.

Entity graph



Entity Graphs are
schemas
populated with
entities

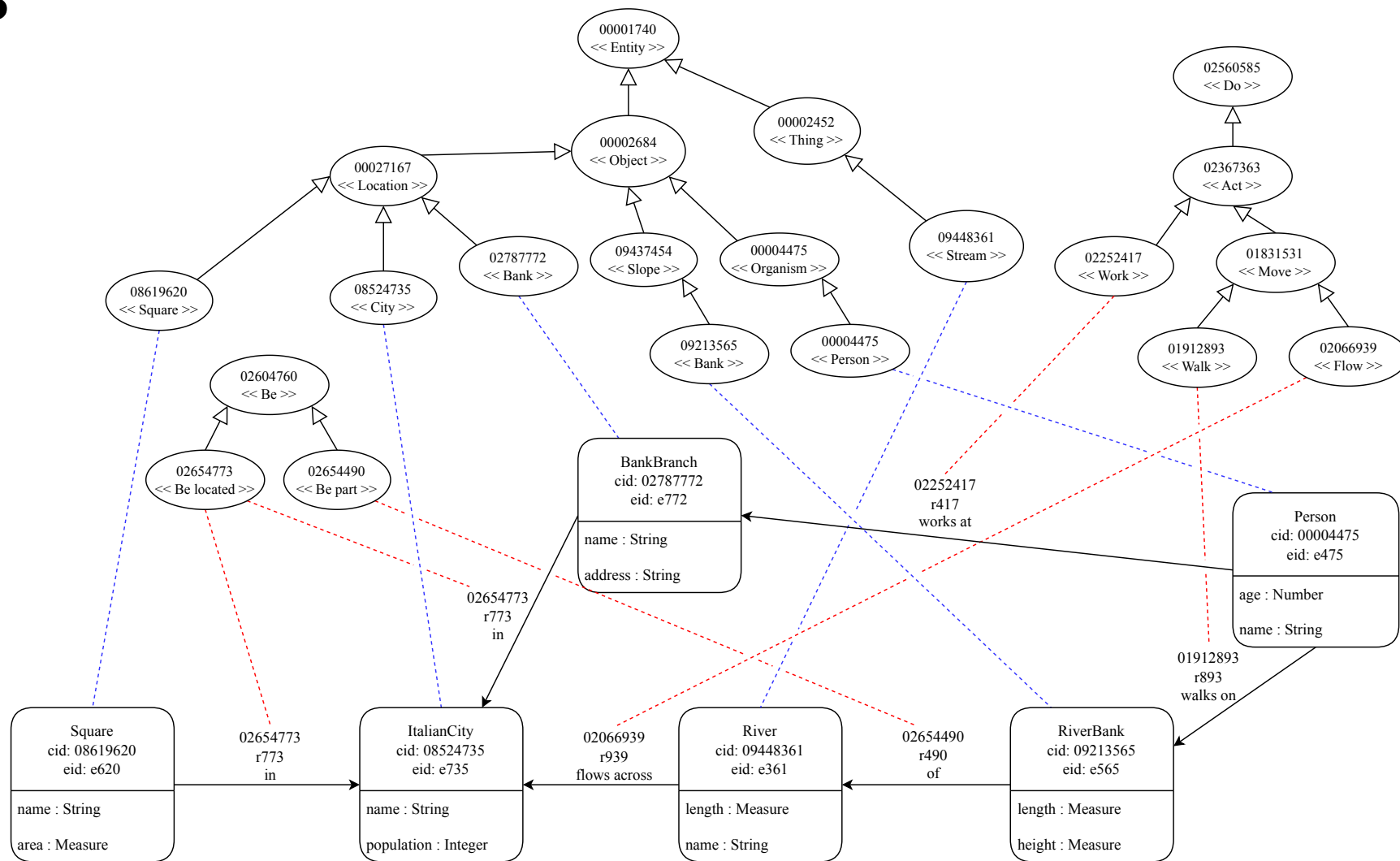
Index

- Concept graphs (CG)
- Entity types and data types
- Etype (entity type) graphs (ETG)
- Entity graphs (EG)
- **Concept + etype graphs**
- Concept + entity graphs
- Etype + entity graphs
- Concept + etype + entity graphs
- Semi-formal world models
- Key notions

World models as concept + etype graphs

Etype graphs
which are complete
with respect to their
reference
concept graphs

The information
represented in these
models cannot be
computed from etype
graphs



Index

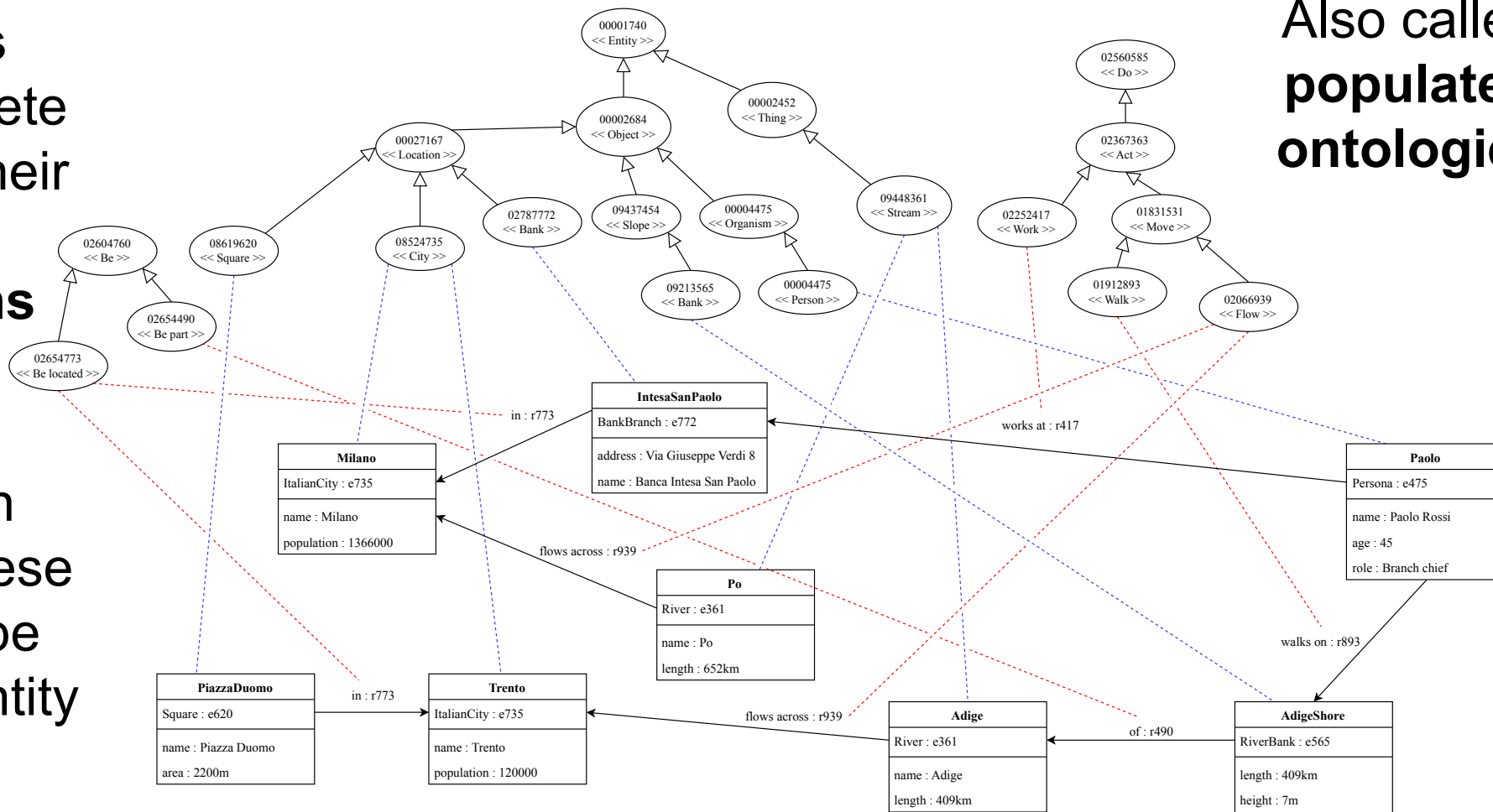
- Concept graphs (CG)
- Etype (entity type) graphs (ETG)
- Entity graphs (EG)
- Concept + etype graphs
- **Concept + entity graphs**
- Etype + entity graphs
- Concept + etype + entity graphs
- Semi-formal world models
- Key notions

World models as concept + entity graphs

Entity graphs
which are complete
with respect to their
reference
concept graphs

Also called
**populated
ontologies**

The information
represented in these
models cannot be
computed from entity
models



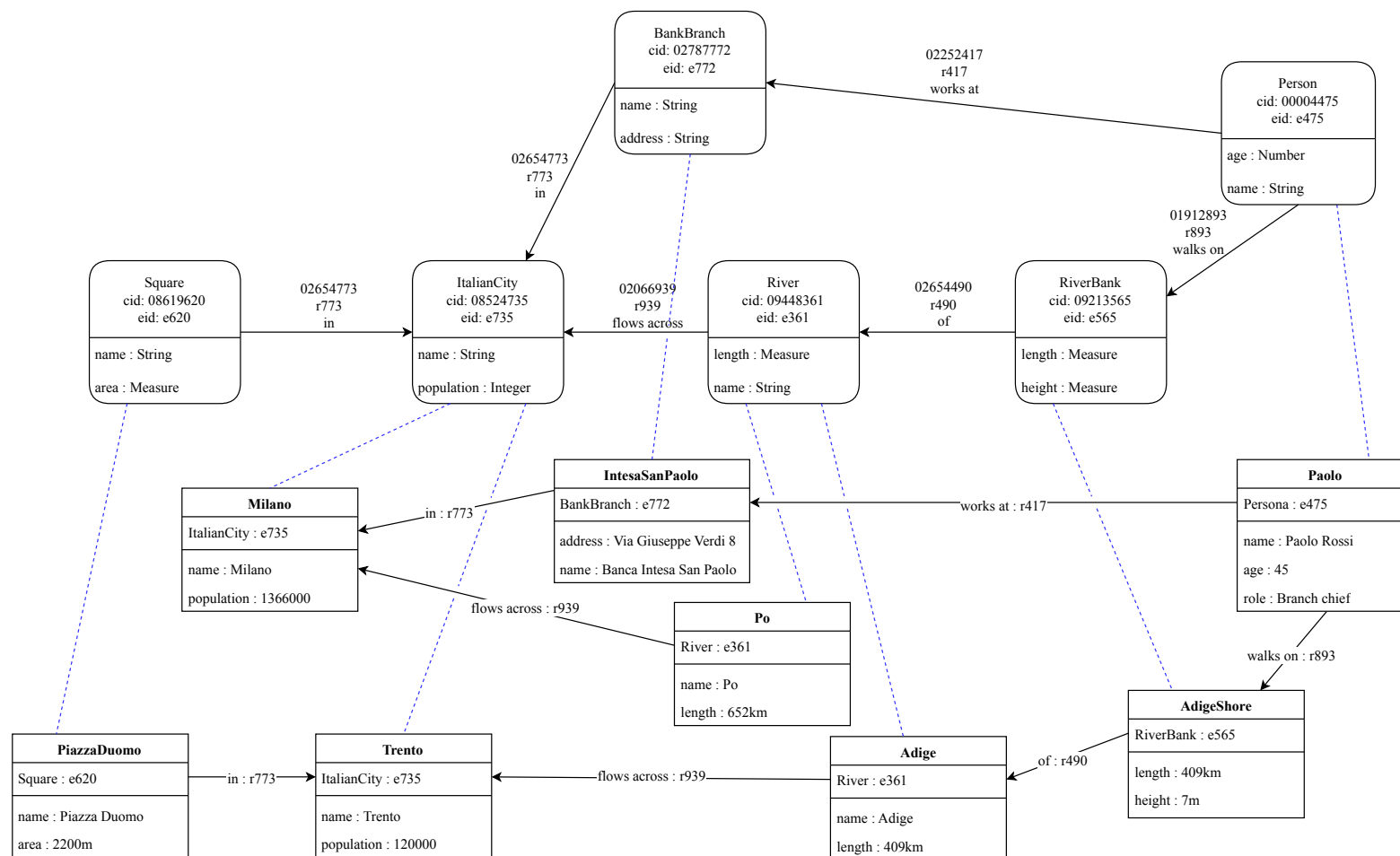
Index

- Concept graphs (CG)
- Etype (entity type) graphs (ETG)
- Entity graphs (EG)
- Concept + etype graphs
- Concept + entity graphs
- **Etype + entity graphs**
- Concept + etype + entity graphs
- Semi-formal world models
- Key notions

World models as etype + entity graphs

entity graphs
which are complete
with respect to
their reference
etype graphs

The information
represented in these
models cannot be
computed from entity
graphs



Index

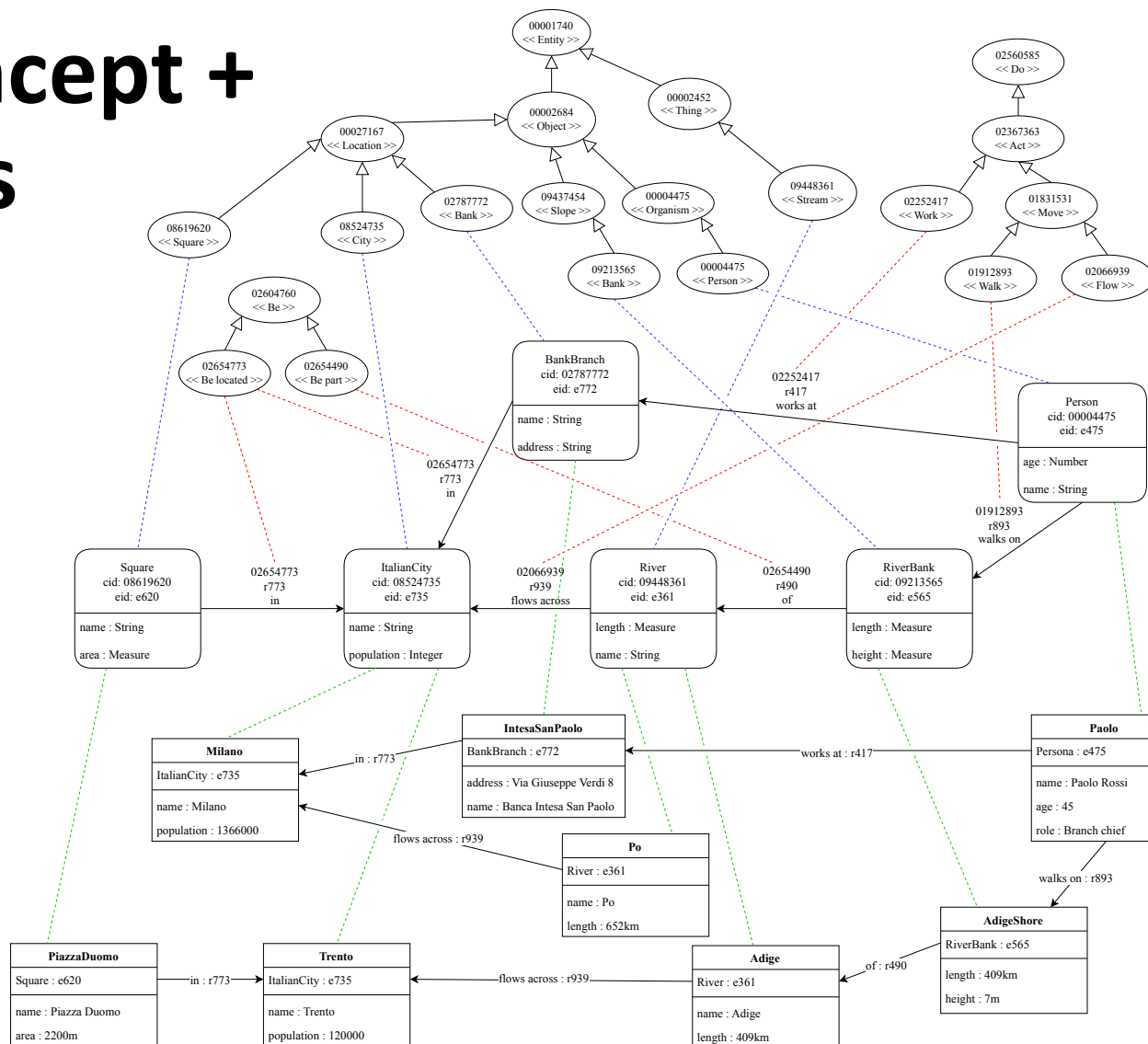
- Concept graphs (CG)
- Entity types and data types
- Etype (entity type) graphs (ETG)
- Entity graphs (EG)
- Concept + etype graphs
- Concept + entity graphs
- Etype + entity graphs
- **Concept + etype + entity graphs**
- Semi-formal world models
- Key notions

World models as concept + etype + entity graphs

entity graphs
which are complete
with respect to their
reference

etype graphs
and **concept graphs**

The information represented in
these models cannot be
computed from etype + entity
graphs



Index

- Concept graphs (CG)
- Entity types and data types
- Etype (entity type) graphs (ETG)
- Entity graphs (EG)
- Concept + etype graphs
- Concept + entity graphs
- Etype + entity graphs
- Concept + etype + entity graphs
- **Semi-formal world models**
- Key notions

Semi-formal canonical world representations

Observation (Canonical world models and representations). The notions of a theory being a theory of a model, and vice versa, of a model being a model of a theory, are meaningful only when the reference world model is known. Once the canonical world model has been selected one can start using the current canonical world representation with no ambiguity.

Observation (Semi-formal canonical world representation, semi-formal world model).

Semi-formal models are the closest approximation to canonical world models. We also call them **semi-formal canonical world models**. In semi-formal models the syntax is well defined and the intended canonical model is often implicitly defined. We have the following mappings:

- Natural language (NL) lexicons and domain languages are semi-formal canonical representations of concept graphs
- EER models are semi-formal canonical representations of knowledge teleontologies
- ER models are semi-formal canonical representations of schemas (teleologies)
- DB tables and knowledge graphs are semi-formal canonical representations of entity graphs

NL Lexicon – example

- [S: \(n\) koto](#) (Japanese stringed instrument that resembles a zither; has a rectangular wooden sounding board and usually 13 silk strings that are plucked with the fingers)
 - [direct hypernym](#) / [inherited hypernym](#) / [sister term](#)
 - [S: \(n\) stringed instrument](#) (a musical instrument in which taut strings provide the source of sound)
 - [S: \(n\) musical instrument, instrument](#) (any of various devices or contrivances that can be used to produce musical tones or sounds)
 - [S: \(n\) device](#) (an instrumentality invented for a particular purpose) *"the device is small enough to wear on your wrist"; "a device intended to conserve water"*
 - [S: \(n\) instrumentality, instrumentation](#) (an artifact (or system of artifacts) that is instrumental in accomplishing some end)
 - [S: \(n\) artifact, artefact](#) (a man-made object taken as a whole)
 - [S: \(n\) whole, unit](#) (an assemblage of parts that is regarded as a single entity) *"how big is that part compared to the whole?"; "the team is a unit"*
 - [S: \(n\) object, physical object](#) (a tangible and visible entity; an entity that can cast a shadow) *"it was full of rackets, balls and other objects"*
 - [S: \(n\) physical entity](#) (an entity that has physical existence)
 - [S: \(n\) entity](#) (that which is perceived or known or inferred to have its own distinct existence (living or nonliving))

On the left hand side is the WordNet lexical hierarchy generalizing the concept for Koto. See [Princeton WordNet](#).
WordNet is a semi-formal world model

S(n) indicates a **synset** associated to a word (here Koto) (one of the possibly many) of synonymous nouns. Here Koto has no synonyms.

Hyponym / Hypernym stand for subclass and superclass relationship .

Each synset is described by a **gloss** (a definition of the meaning of a word, most of the time incomplete, provided informally) and an **example** (between quotes in the figure on the left).

NL Lexicon as a semi-formal ontology

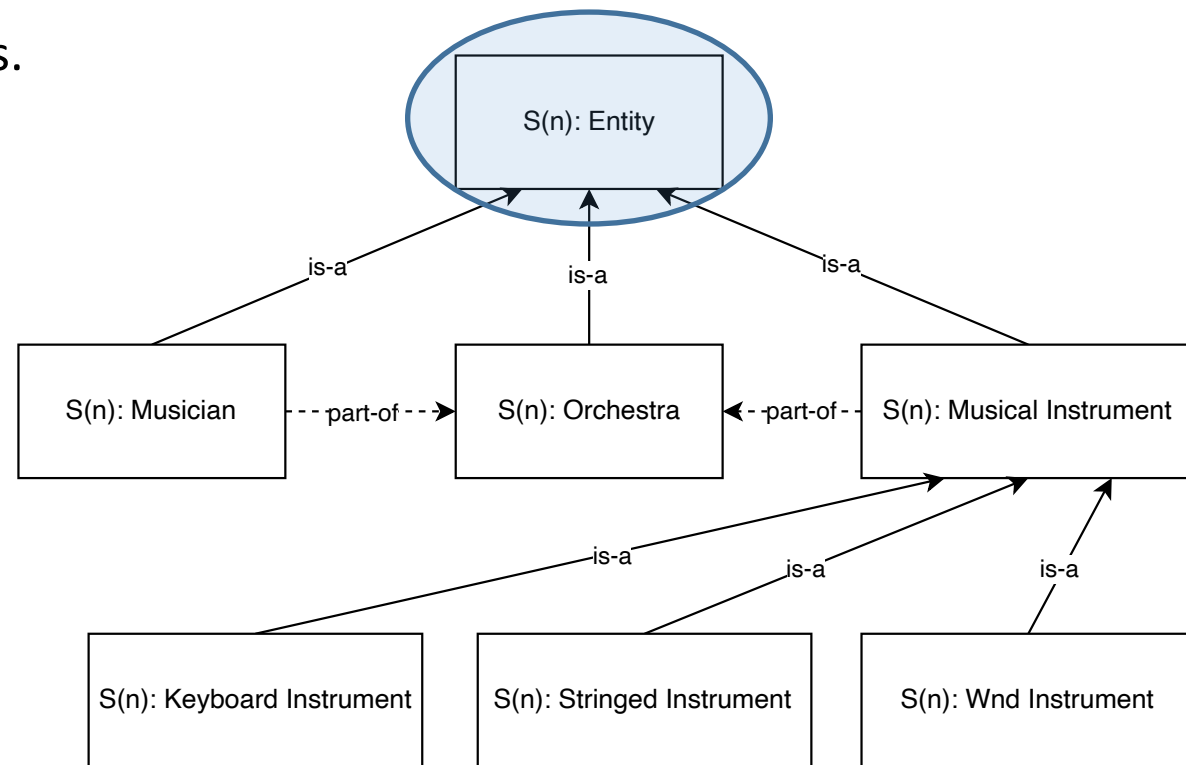
Lexicons. Lexicons are lists of words which encode the meaning of the **words** of one or more natural languages. Lexicons adhere to **high quality principles**. They are usually developed or validated by humans

Words, as they occur in languages, are the main elements of lexicons.

Senses. A sense is one of the multiple **meanings** denoted by a polysemous word (e.g., *car* stands for *automobile* and *railway car*). The word **sense** can be taken as synonymous of the word **concept** (see above).

Synsets. Sets of **synonyms**, that is, synonymous words associated to the same sense.

Unique identifiers (not visible to users) named *GID*, e.g., 588967, are the names denoting concepts (via the data type concept).



Domain Lexicon as a semi-formal domain LTLO

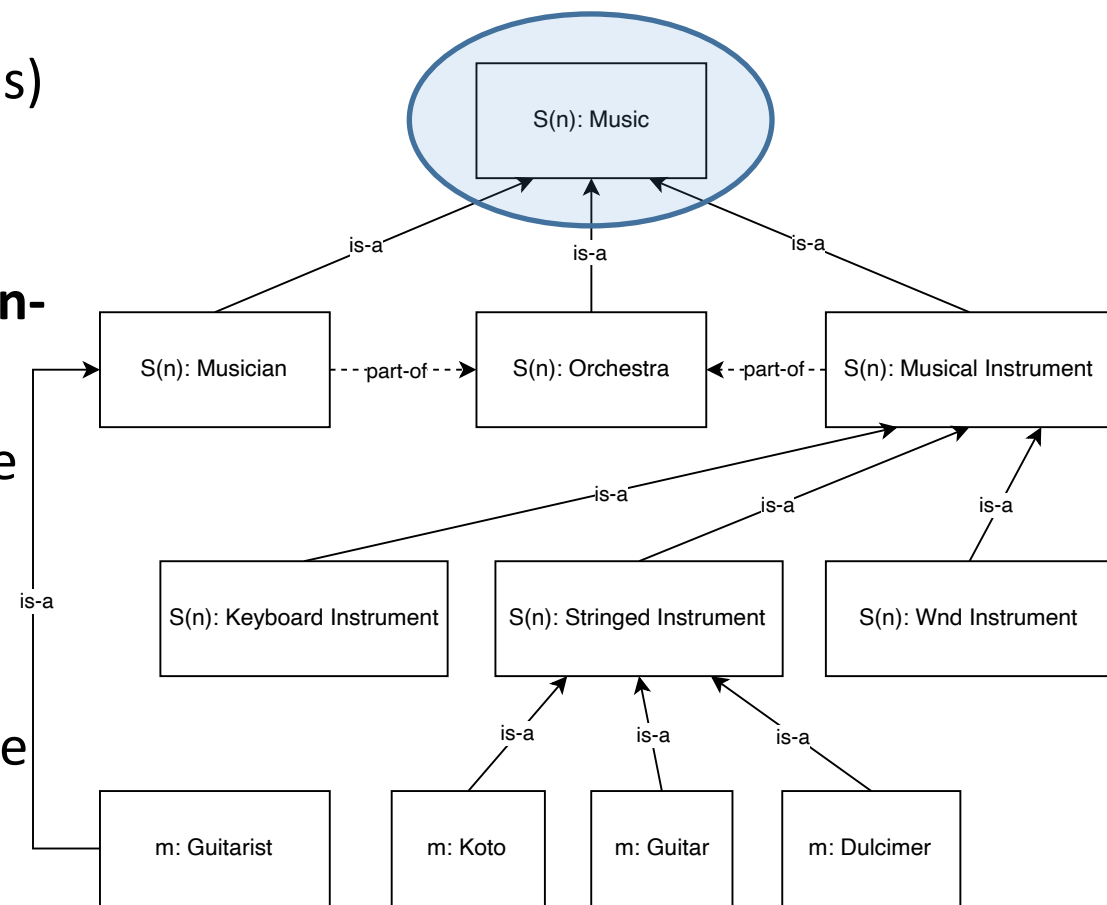
Domain Lexicons (e.g., **Standards**, or **name spaces**) are lexicons which encode the semantics of domain language(s) which focus on specific application domains (e.g., health, tourism). **Extensively used in the Web and in standards**

They usually extend natural language lexicons with **domain-specific words** and **concepts**.

Words in domain languages follow the same rules as those in natural languages with one exception: **they are NOT polysemous**, i.e., they have only one sense.

Each word has a **prefix**, e.g., *mi:* (e.g., for musical instruments) to indicate the domain language/ name space to which the words is associated.

As with lexicons, each sense is identified via a **unique identifier** named *GID*, e.g., 5667853 for *mi:Koto*, denoting the single sense of that word.



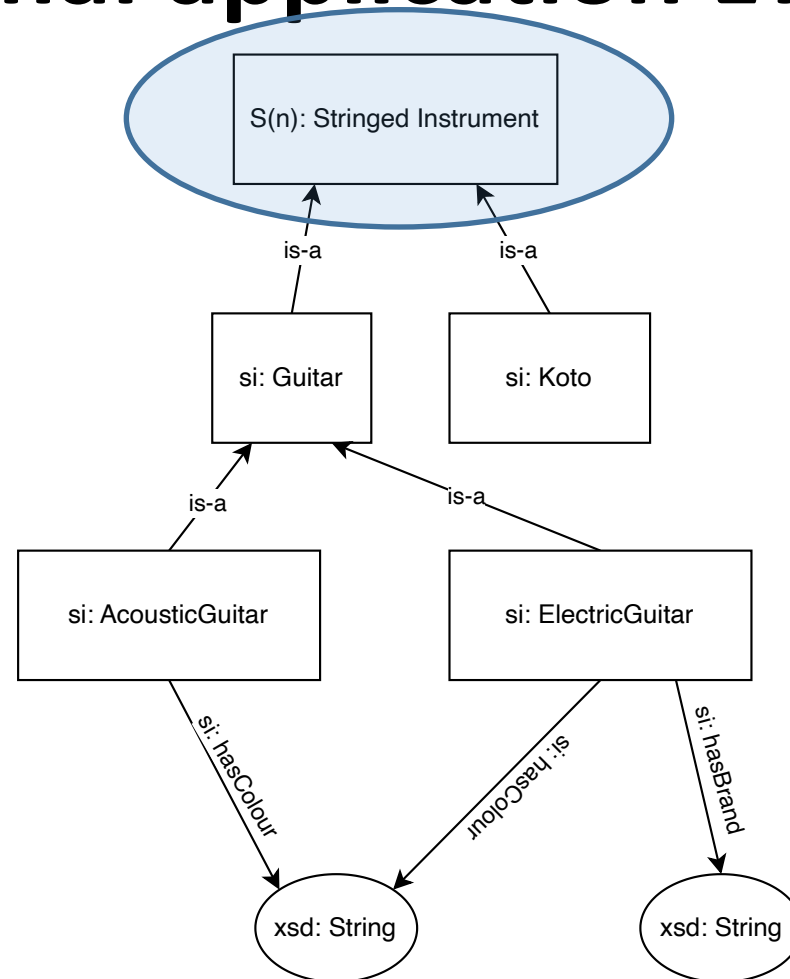
Application Lexicon as a semi-formal application LTLO

Application lexicons are specialized lexicons which focus on a purpose-specific (application dependent) part of a lexicon. They are usually built by selecting concepts from domain (or natural language) lexicons.

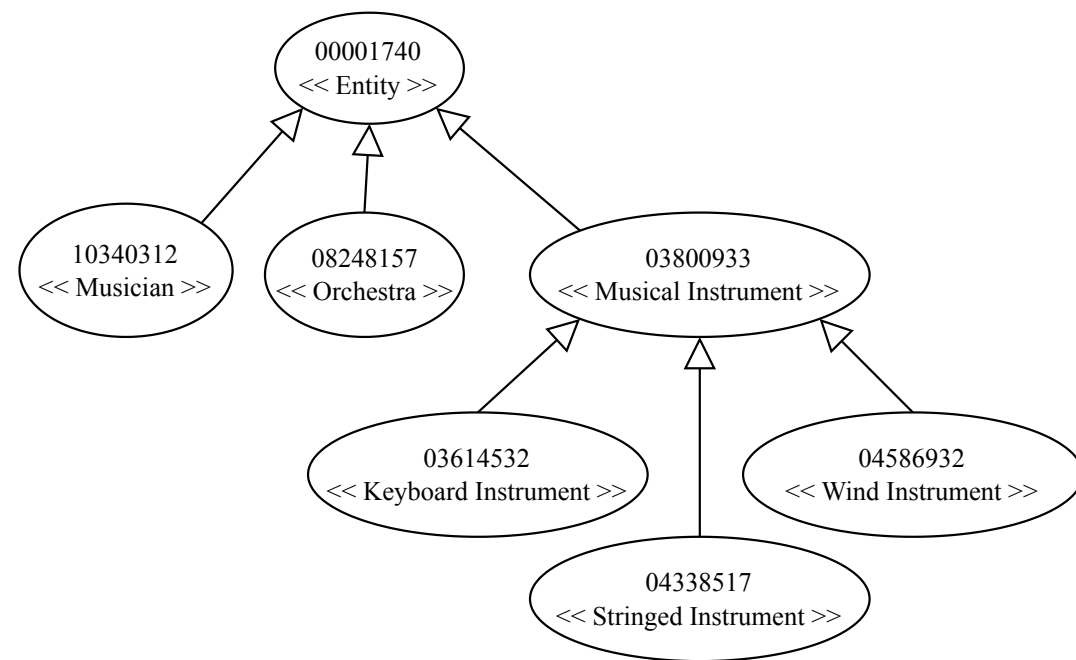
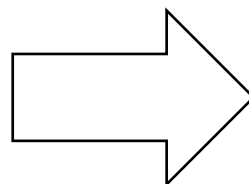
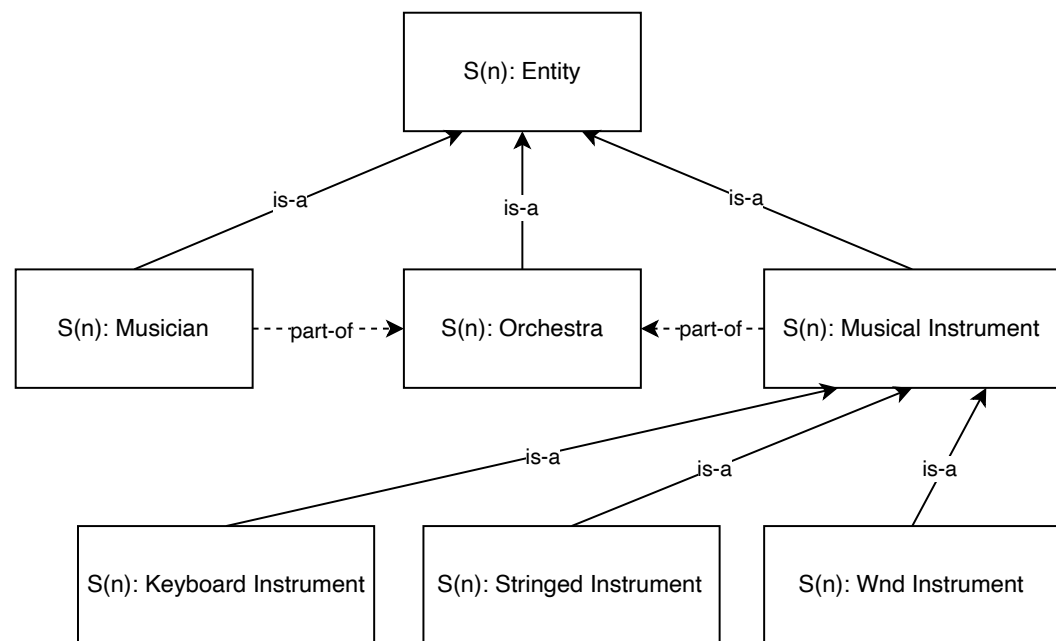
Application lexicons are obtained from lexicons by:

- Identifying the root concept, as from the specific need. This is a whole defining the reference space and time containment. In the example above: orchestra
- Name the root concept as «object» (or «thing», or «entity» or anything more specific, e.g., «musical instrument», to define the scope of the concepts which are selected
- Keeping the concepts of all the objects relevant to the application
- Keeping the relevant concepts which specialize, via the IS-A hierarchy, the concepts from previous step
- Dropping irrelevant concepts (below/ above the whole)

Observation (Application lexicons). Application lexicons are the lexicons used in ER/ EER models. Most often the words used are in natural language and therefore polysemous



From a semi-formal CG to an ontology - example



Natural Language (NL) Lexicon – example (recall)

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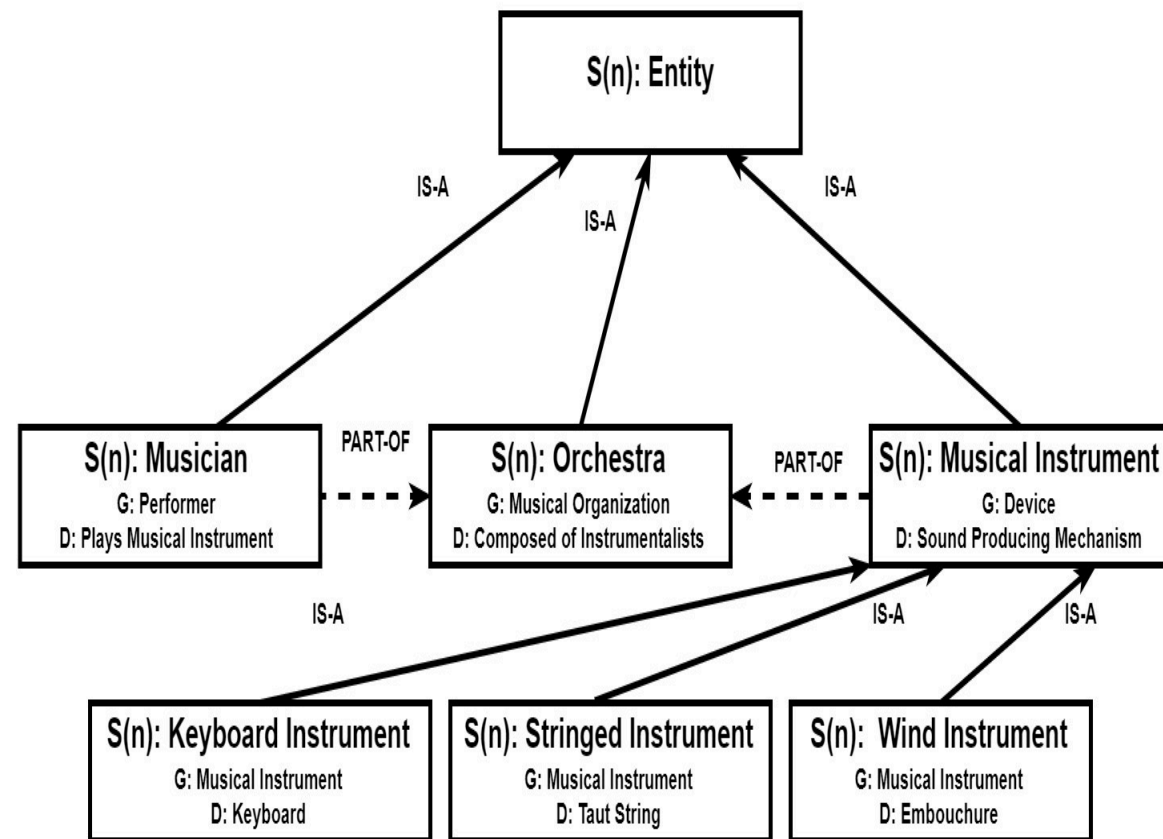
NL Lexicon as a semi-formal KTLO

A lexicon is a semi-formal **IS-A /MG hierarchy** which semantically models superclass - subclass (hyponym / hypernym) relations between senses. **Any node is both a concept and an etype**

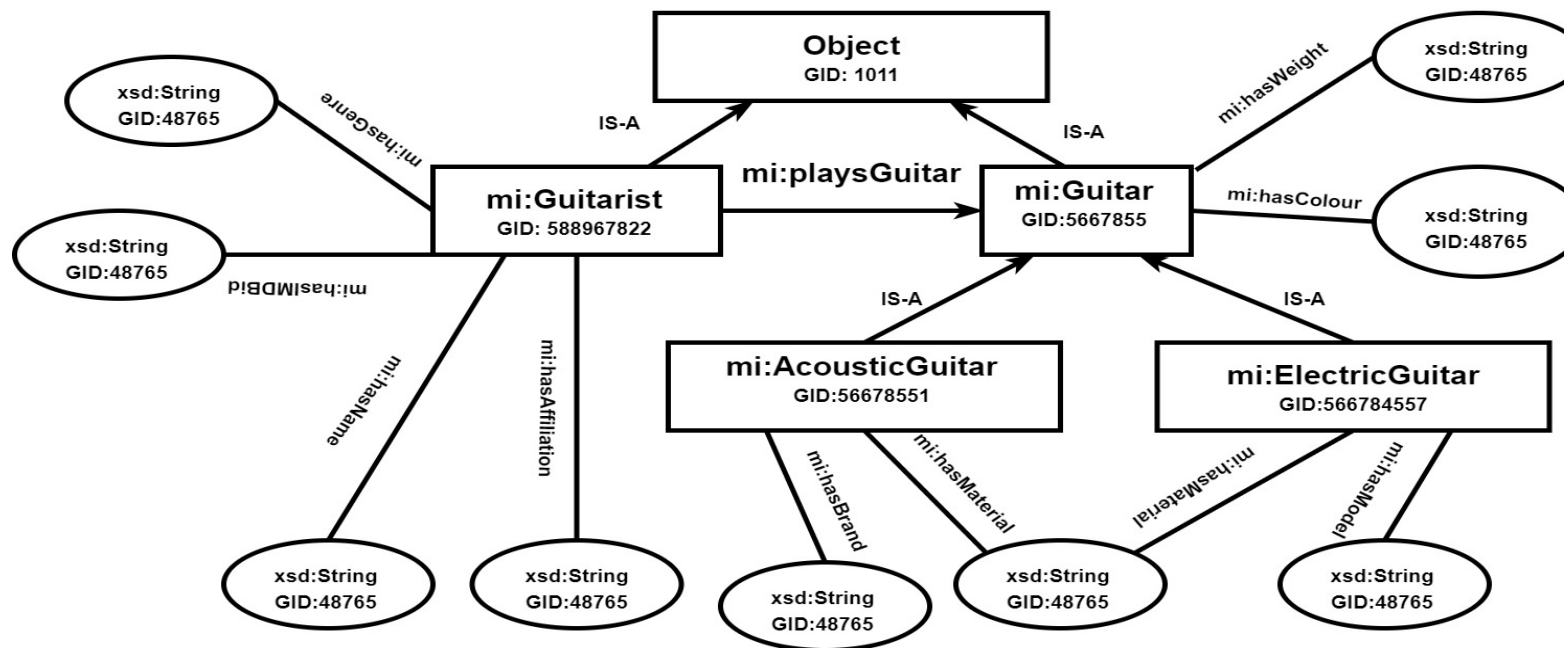
Genus-Differentia is the main means by which the IS-A hierarchy are built

Genus. The set of properties which define the scope of a sense, e.g., *musical instrument* (G) for *stringed instrument*. Any concept in a hierarchy has, as genus, the concept of the single parent node.

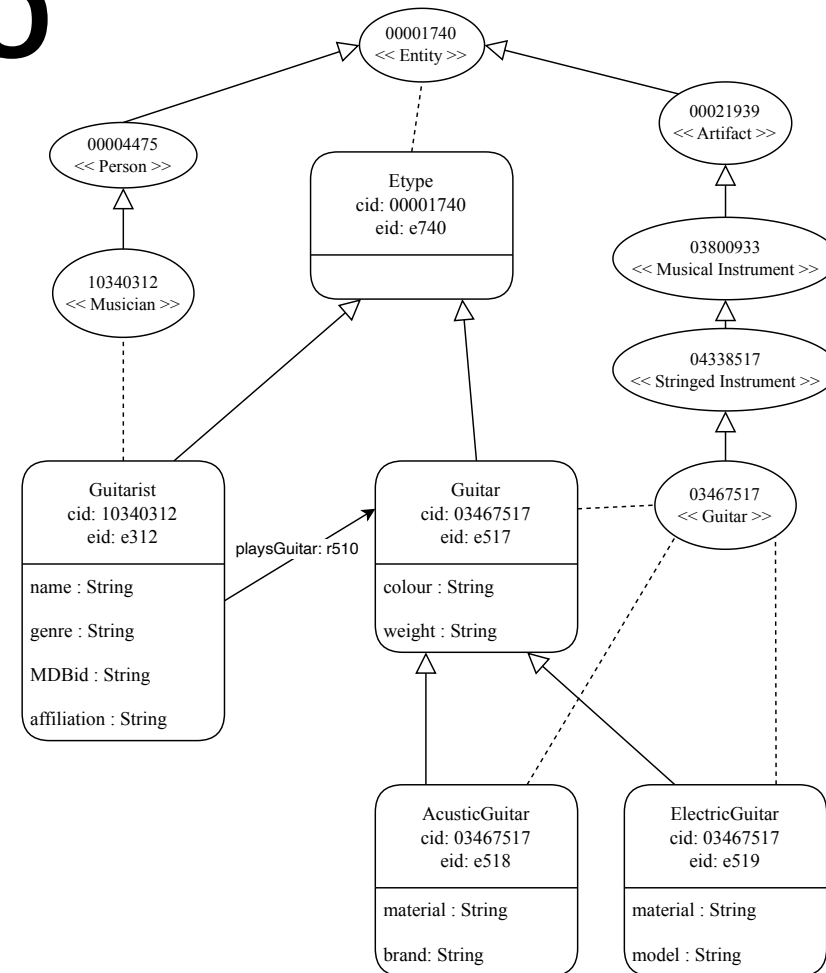
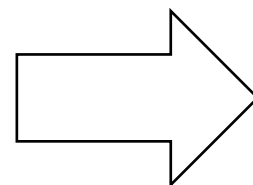
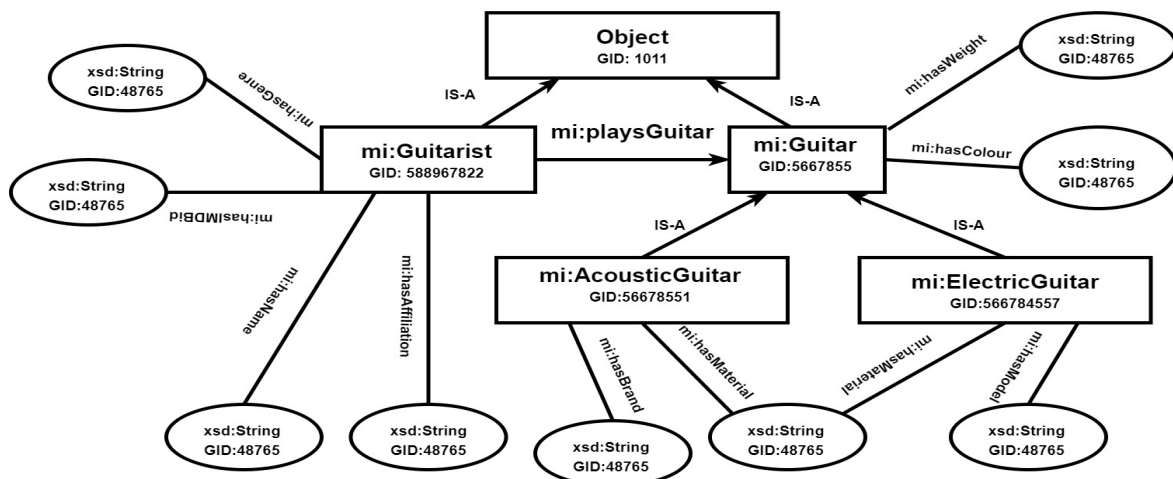
Differentia. The set of properties which qualify / differentiate senses with the same genus, e.g., *Taut String* (D) for *Stringed Instrument*. All the siblings of the same concept, while having the same genus have different differentias. All the differentias are **disjoint**, this guarantees the no ambiguity (of the concepts associated to polysemous) words,



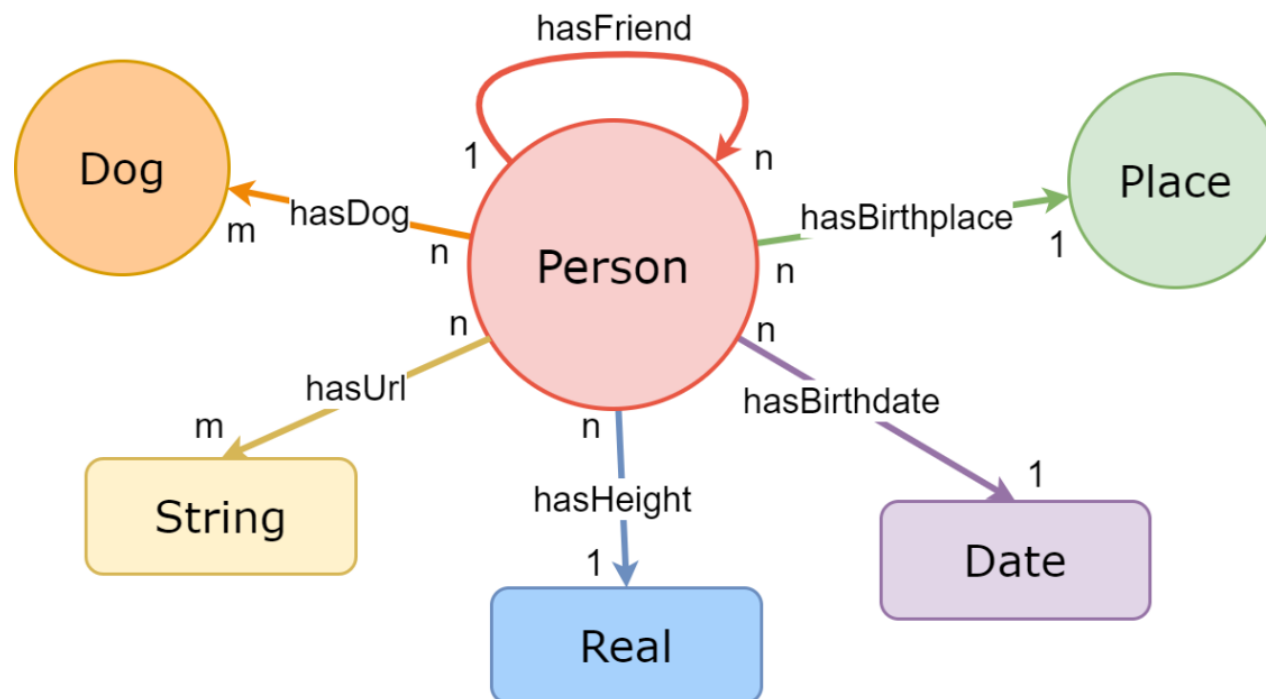
Semi-formal KTLO - example



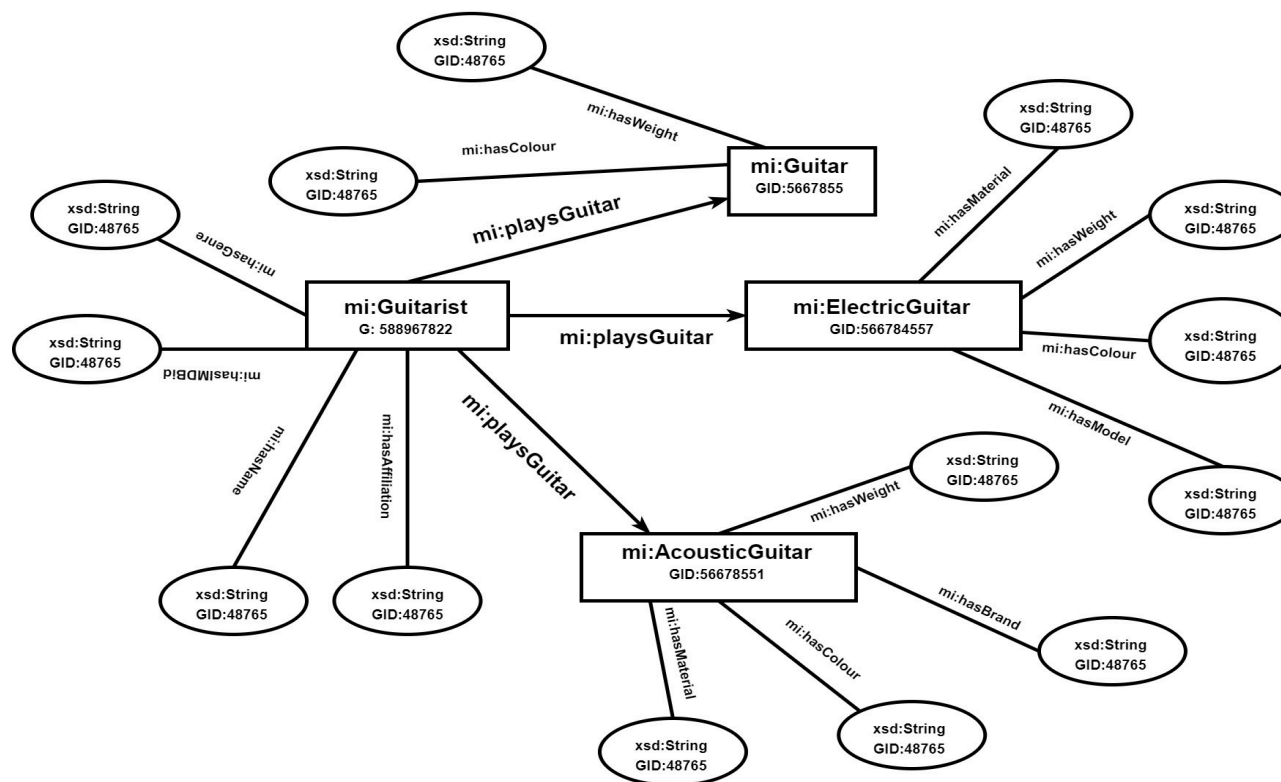
From semi-formal KTLO to KTLO



Semi-formal TLO – example

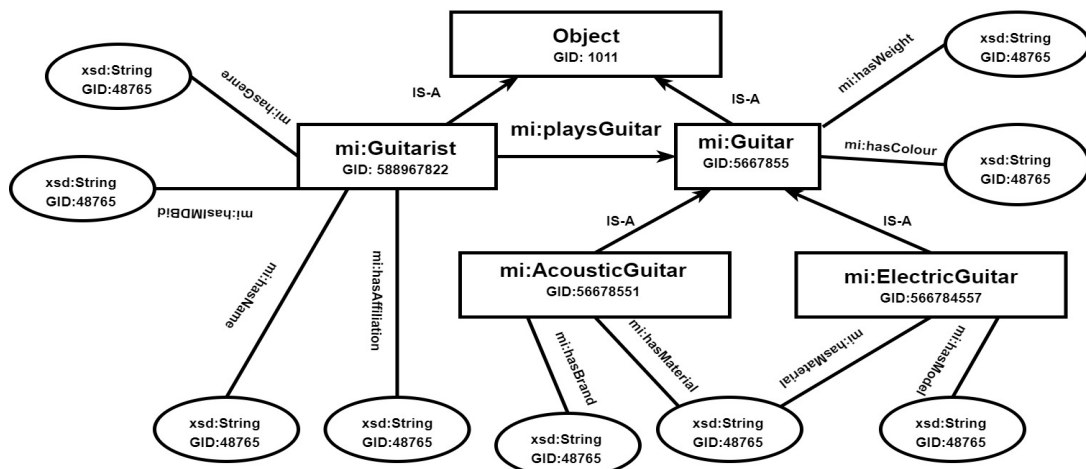


Semi-formal TLO of previous KTLO - example

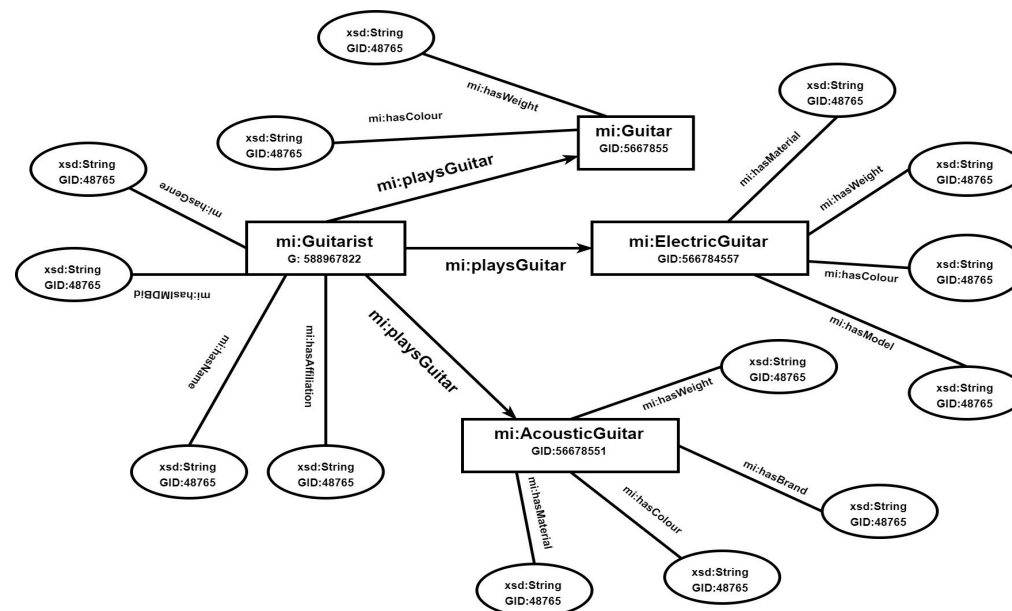


UNFOLDING - from KTLO to TLO – example

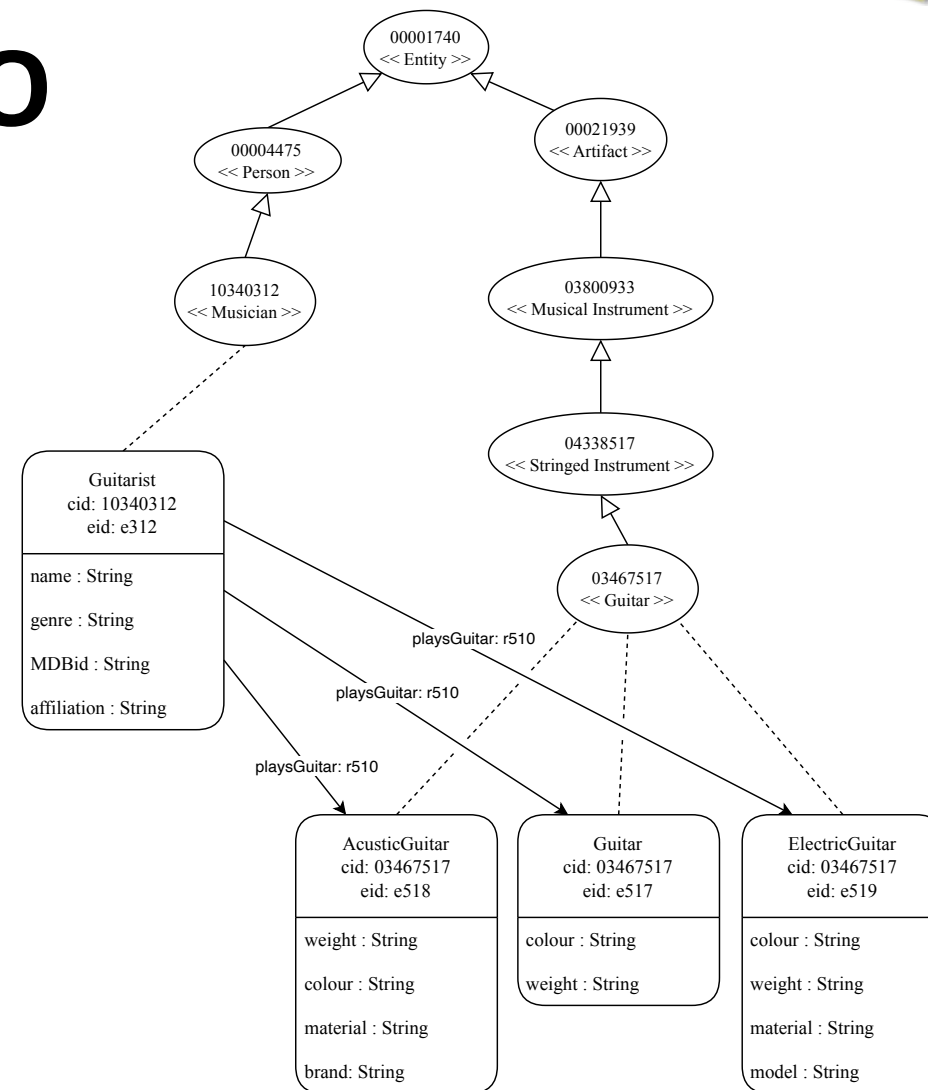
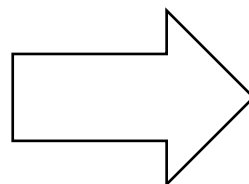
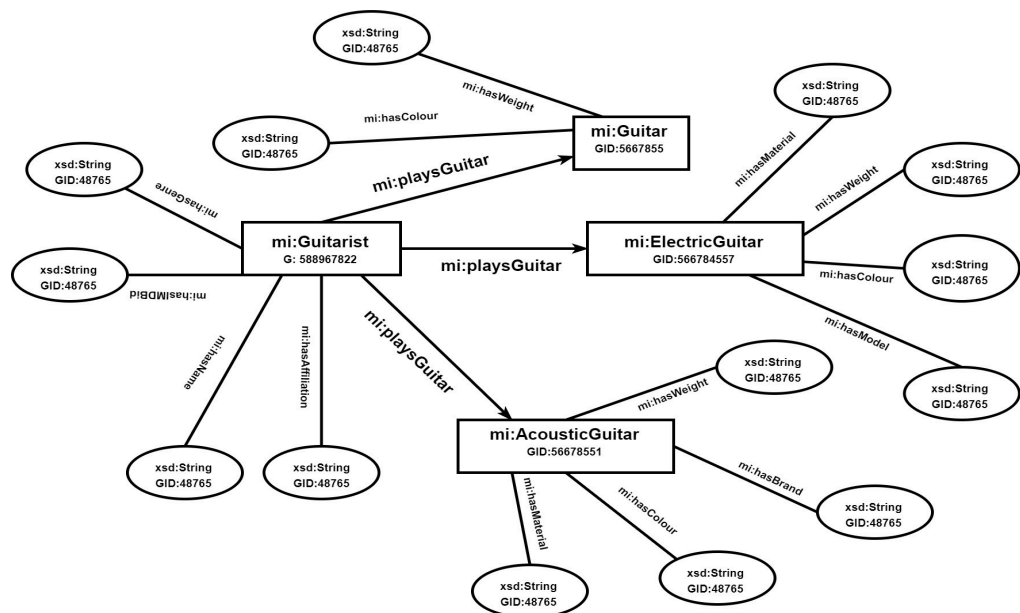
FROM



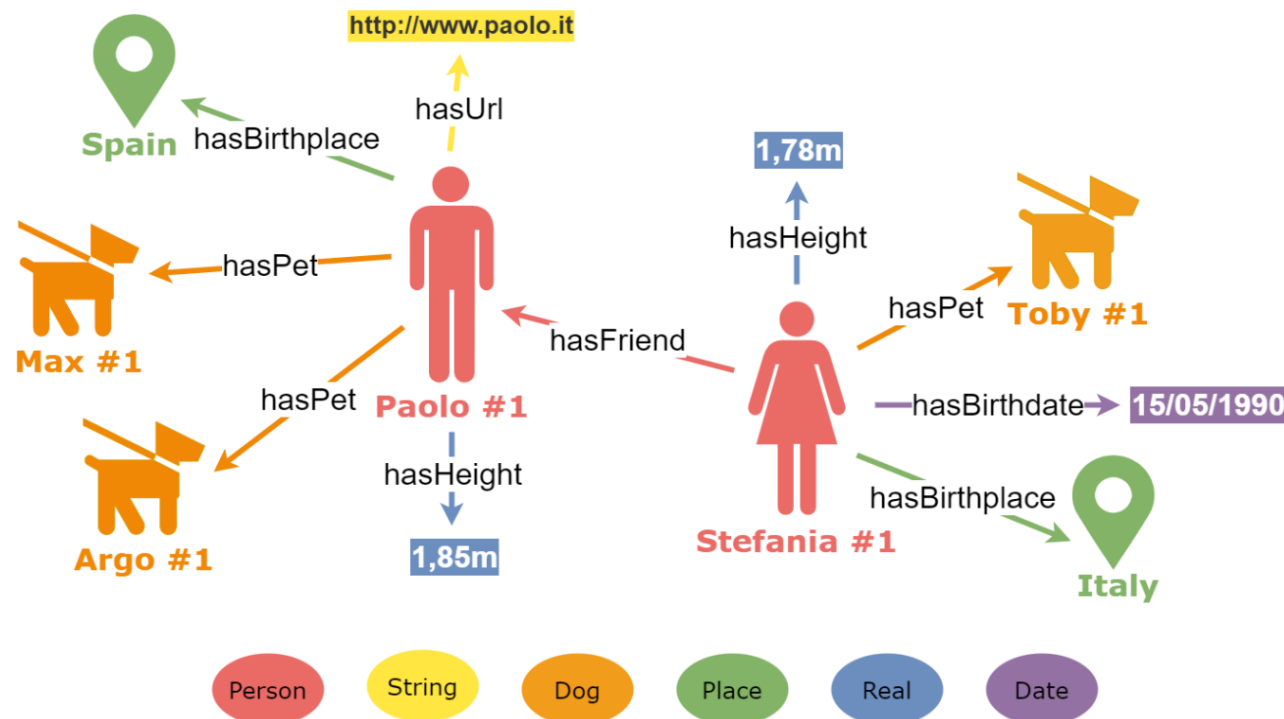
TO



From semi-formal TLO to TLO

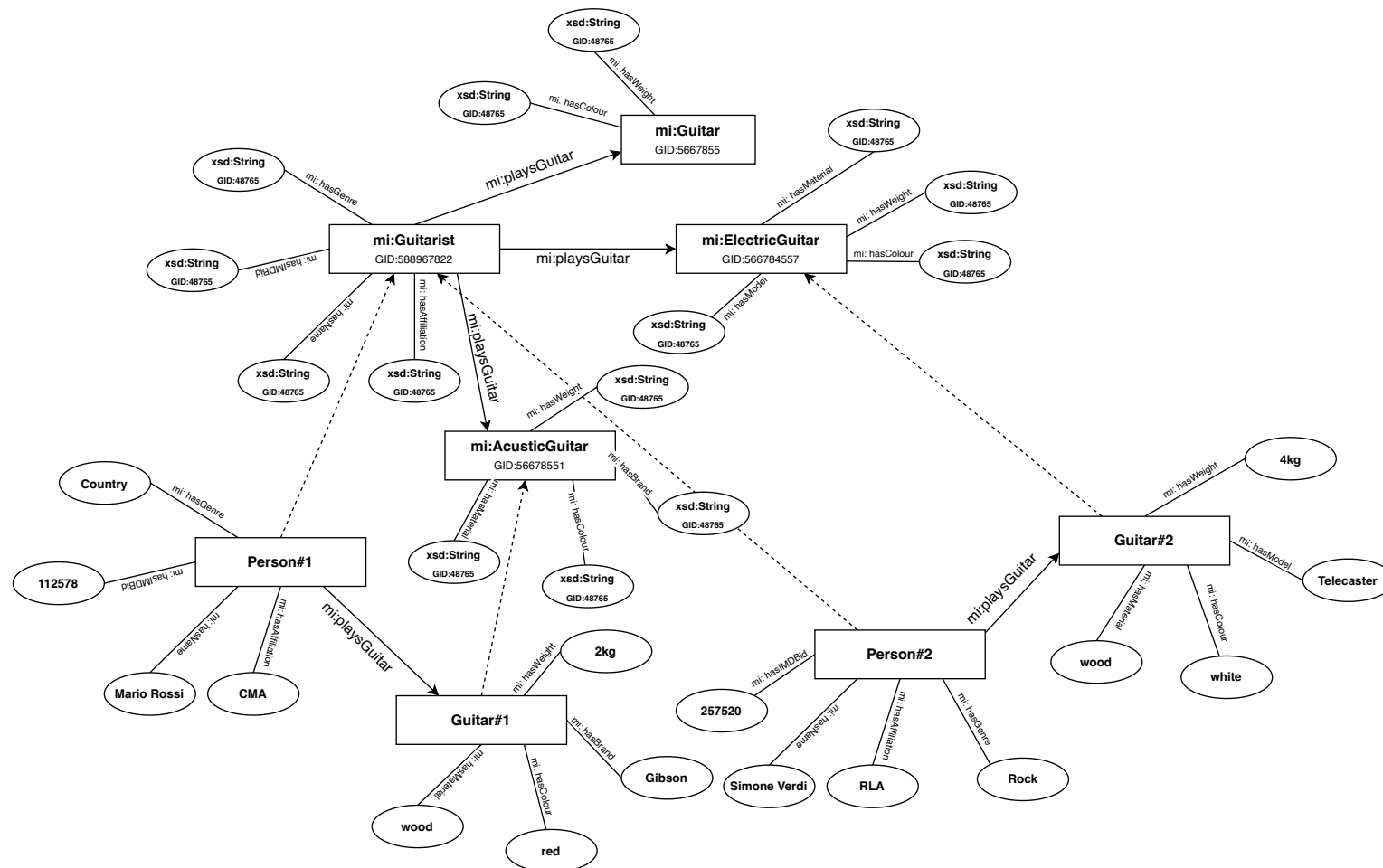


Semi-formal EG of previous TLO - example



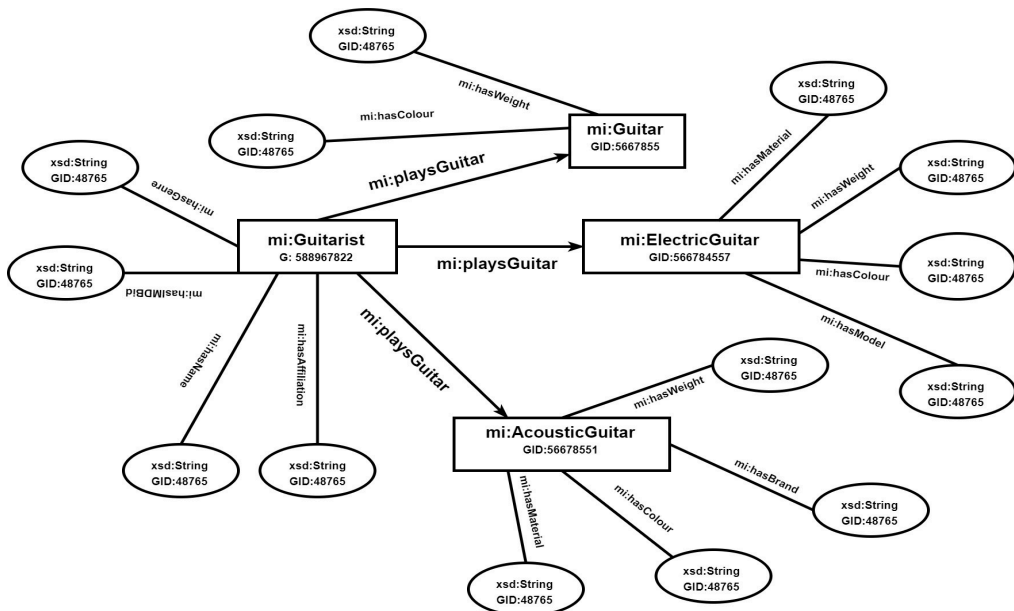
Observation (ETG, EG). An ETG defines all the etypes, dtypes, object properties and attributes used in an EG. An EG is an expansion (notion formally defined later on) where each node and link is expanded into all its elements. Compare with the previous ETG.

Semi-formal EG of previous TLO - example

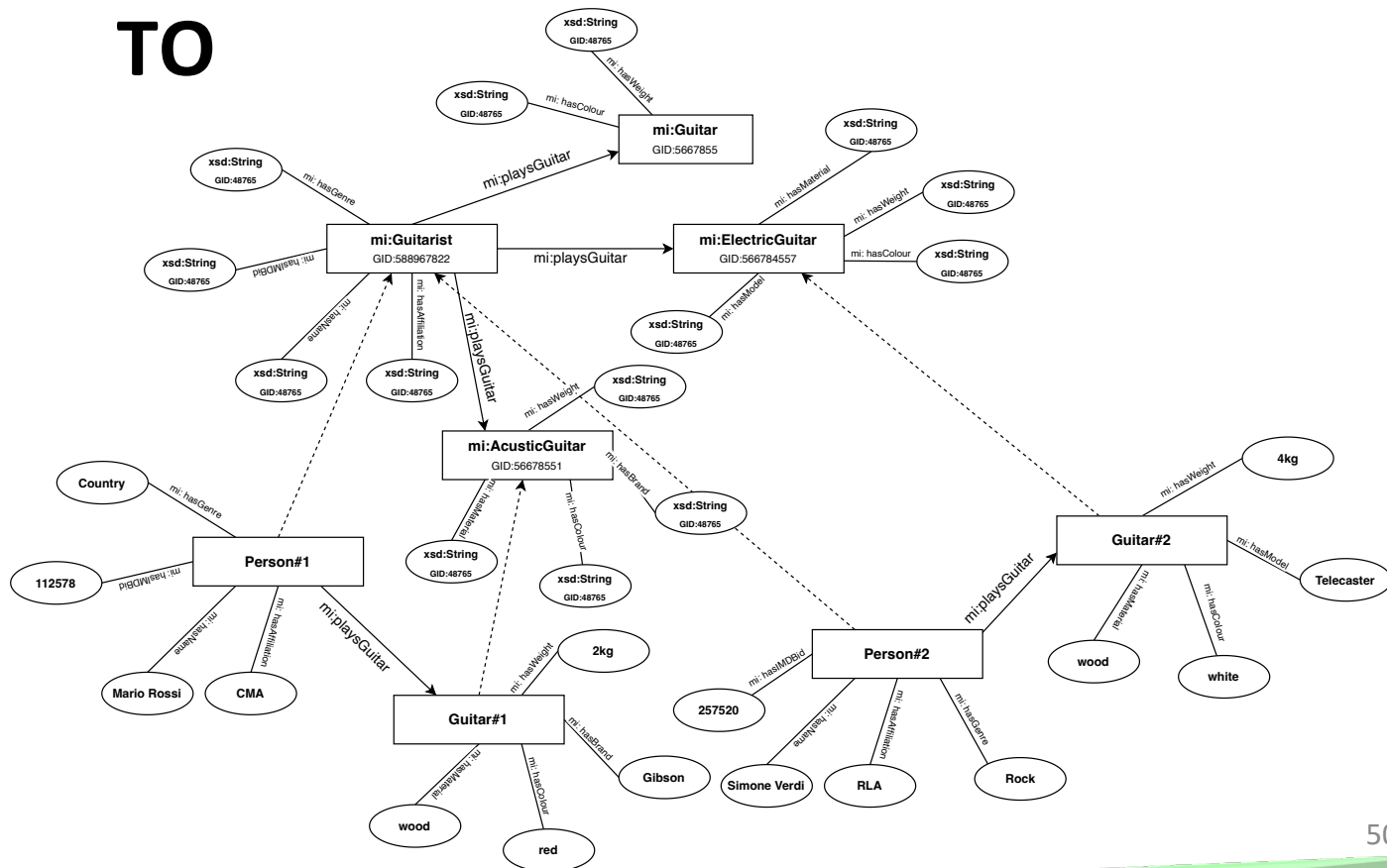


EXPANSION - from TLO to EG – example

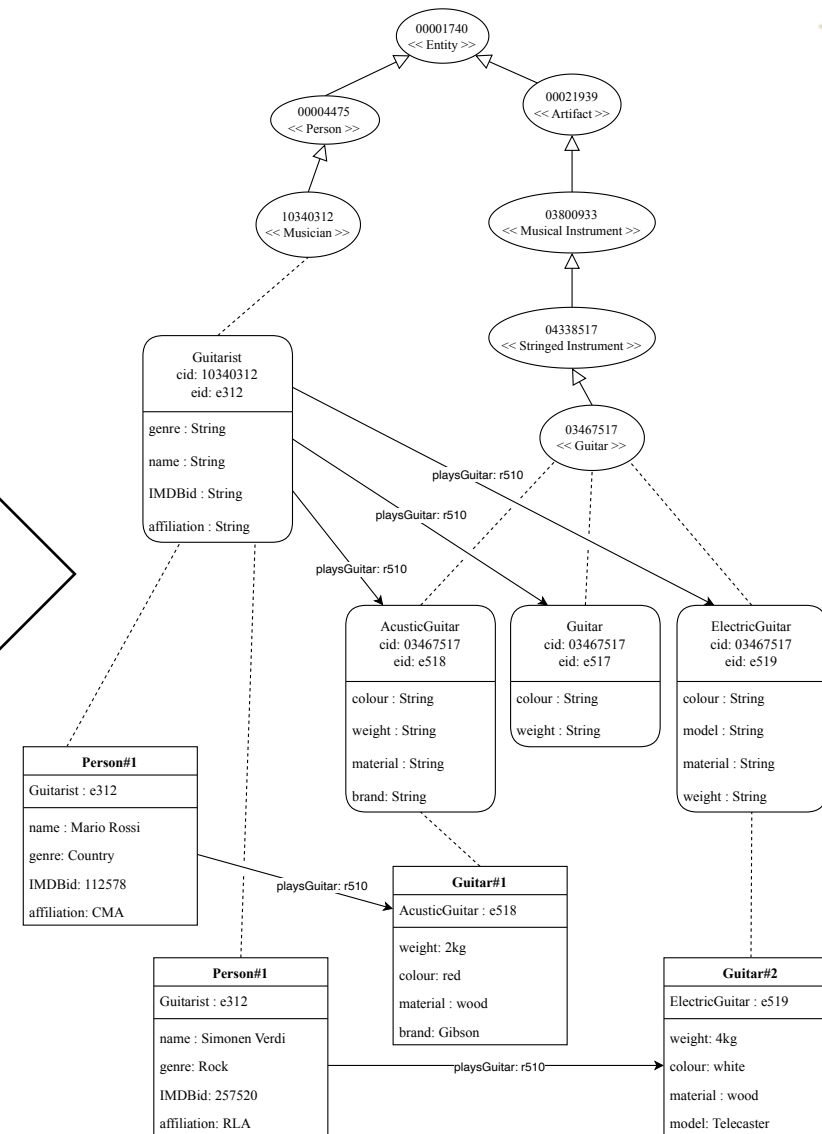
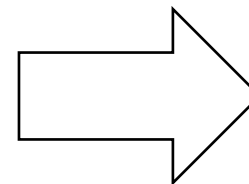
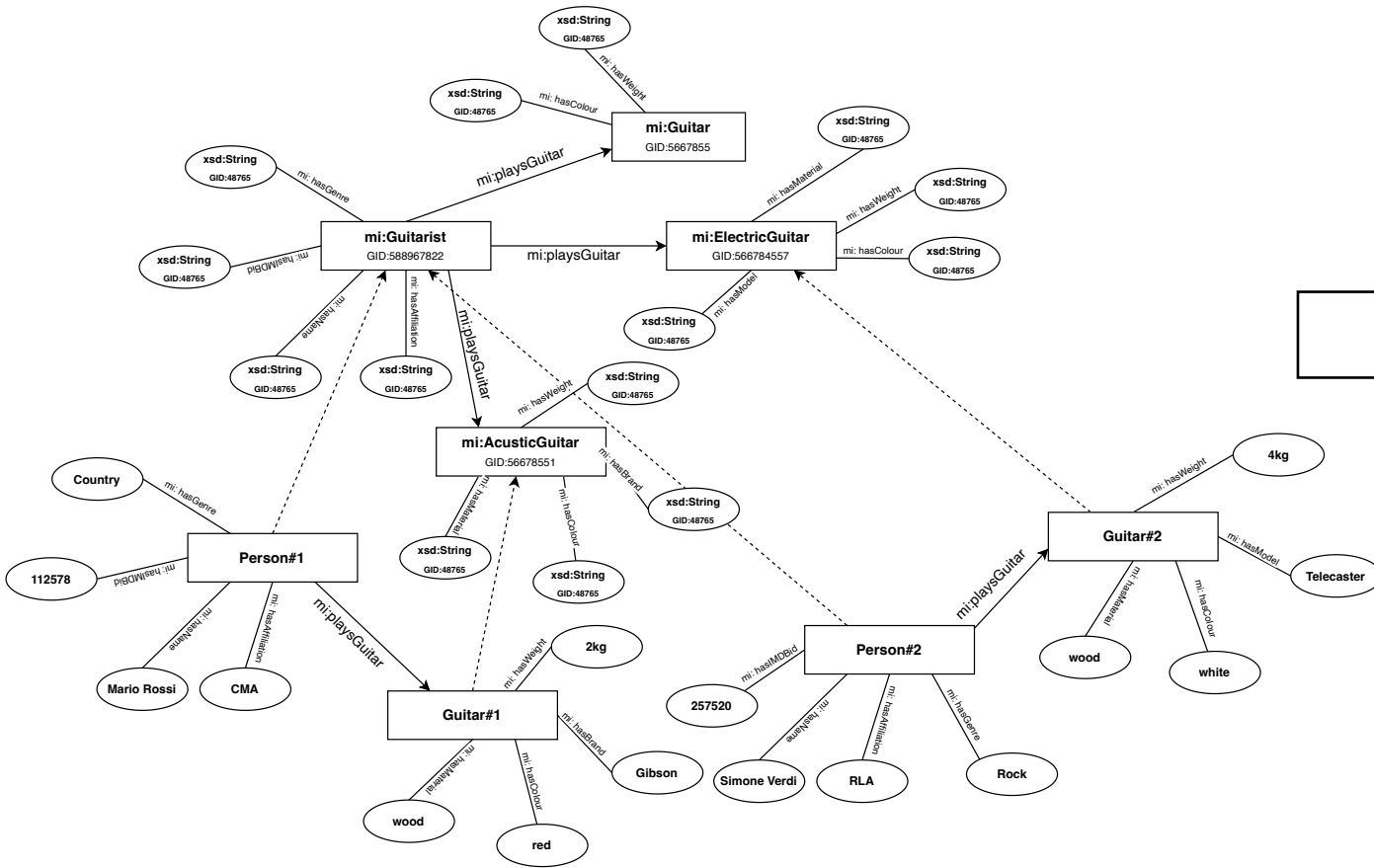
FROM



TO



From semi-formal EG to EG



Semi-formal vs. (canonical) world models

Observation (Base and composed formal world models). We have provided 3 types of base world models further differentiated depending on (i) the level of generality of the root node and, in the case of etype graphs also, on (ii) the nesting of etypes with respect to the root concept.

Composed world models are obtained from base world model by making explicit the ISA and MG/LG hierarchies of concepts and etypes, respectively. As written above, differently from base world models, composed world models are complete representations of the relations holding among concepts, etypes and entities.

Observation (Semi-formal and formal world models). Semi-formal world models are ambiguous representations of formal base world models of various kinds. There are two possible types of mistakes:

- Ambiguity of the words used (words instead of concepts). **This problem is solved by canonical world models.**
- Incompleteness of the semi-formal world model (as it relates to the incompleteness of formal base models). This problem cannot be solved by world models (e.g., Wordnet as 100+ thousand nodes and millions of links). The unfolding of etypes into more specific etypes must be done dynamically and purpose driven. **This problem is solved by world logics, via a formalization and algorithmic implementation of unfolding and expansion.**

Index

- Concept graphs (CG)
- Entity types and data types
- Etype (entity type) graphs (ETG)
- Entity graphs (EG)
- Concept + etype graphs
- Concept + entity graphs
- Etype + entity graphs
- Concept + etype + entity graphs
- **Key notions**

Key notions

- Etypes, data types
- Object properties, data properties
- Concept graphs (CG)
- Ontologies , populated ontologies
- Language teleontologies (LTLO)
- Etype graphs (ETG)
- Knowledge Teleontologies (KTLO)
- Knowledge Taxonomies (KTAX)
- Teleologies (TLO)
- Schemas, schema graphs (SG)
- Entity graphs (EG), populated schemas
- Composed world models



World model Types (HP2T)